paul graham

on society

essays, volume seven

printed at home, october 2026

13 essays, 42,064 words. all of them are at paulgraham.com, free, as they should be. this is one reader's copy, printed at home to be read away from a screen, with a pencil.

a printed book of essays
github.com/workinprogmess/a-printed-book-of-essays

contents

01	what you can't say	january 2004	4
02	inequality and risk	august 2005	14
03	economic inequality	january 2016	20
04	modeling a wealth tax	august 2020	28
05	how to convert between wealth and income tax	may 2026	30
06	the refragmentation	january 2016	34
07	cities and ambition	may 2008	46
08	the four quadrants of conformism	july 2020	54
09	orthodox privilege	july 2020	60
10	the two kinds of moderate	december 2019	64
11	haters	january 2020	68
12	the origins of wokeness	january 2025	72
13	the brand age	march 2026	84

01

what you can't say

january 2004
paulgraham.com/say.html

have you ever seen an old photo of yourself and been embarrassed at the way you looked? *did we actually dress like that?* we did. and we had no idea how silly we looked. it's the nature of fashion to be invisible, in the same way the movement of the earth is invisible to all of us riding on it.

what scares me is that there are moral fashions too. they're just as arbitrary, and just as invisible to most people. but they're much more dangerous. fashion is mistaken for good design; moral fashion is mistaken for good. dressing oddly gets you laughed at. violating moral fashions can get you fired, ostracized, imprisoned, or even killed.

if you could travel back in a time machine, one thing would be true no matter where you went: you'd have to watch what you said. opinions we consider harmless could have gotten you in big trouble. i've already said at least one thing that would have gotten me in big trouble in most of europe in the seventeenth century, and did get galileo in big trouble when he said it — that the earth moves.[1]

it seems to be a constant throughout history: in every period, people believed things that were just ridiculous, and believed them so strongly that you would have gotten in terrible trouble for saying otherwise.

is our time any different? to anyone who has read any amount of history, the answer is almost certainly no. it would be a remarkable coincidence if ours were the first era to get everything just right.

it's tantalizing to think we believe things that people in the future will find ridiculous. what *would* someone coming back to visit us in a time machine have to be careful not to say? that's what i want to study here. but i want to do more than just shock everyone with the heresy du jour. i want to find general recipes for discovering what you can't say, in any era.

the conformist test

let's start with a test: do you have any opinions that you would be reluctant to express in front of a group of your peers?

if the answer is no, you might want to stop and think about that. if everything you believe is something you're supposed to believe, could that possibly be a coincidence? odds are it isn't. odds are you just think what you're told.

the other alternative would be that you independently considered every question and came up with the exact same answers that are now considered acceptable. that seems unlikely, because you'd also have to make the same mistakes. mapmakers deliberately put slight mistakes in their maps so they can tell when someone copies them. if another map has the same mistake, that's very convincing evidence.

like every other era in history, our moral map almost certainly contains a few mistakes. and anyone who makes the same mistakes probably didn't do it by accident. it would be like someone claiming they had independently decided in 1972 that bell-bottom jeans were a good idea.

if you believe everything you're supposed to now, how can you be sure you wouldn't also have believed everything you were supposed to if you had grown up among the plantation owners of the pre-civil war south, or in germany in the 1930s — or among the mongols in 1200, for that matter? odds are you would have.

back in the era of terms like "well-adjusted," the idea seemed to be that there was something wrong with you if you thought things you didn't dare say out loud. this seems backward. almost certainly, there is something wrong with you if you *don't* think things you don't dare say out loud.

trouble

what can't we say? one way to find these ideas is simply to look at things people do say, and get in trouble for.[2]

of course, we're not just looking for things we can't say. we're looking for things we can't say that are true, or at least have enough chance of being true that the question should remain open. but many of the things people get in trouble for saying probably do make it over this second, lower threshold. no one gets in trouble for saying that 2 + 2 is 5, or that people in pittsburgh are ten feet tall. such obviously false statements might be treated as jokes, or at worst as evidence of insanity, but they are not likely to make anyone mad. the statements that make people mad are the ones they worry might be believed. i suspect the statements that make people maddest are those they worry might be true.

if galileo had said that people in padua were ten feet tall, he would have been regarded as a harmless eccentric. saying the earth orbited the sun was another matter. the church knew this would set people thinking.

certainly, as we look back on the past, this rule of thumb works well. a lot of the statements people got in trouble for seem harmless now. so it's likely that visitors from the future would agree with at least some of the statements that get people in trouble today. do we have no galileos? not likely.

to find them, keep track of opinions that get people in trouble, and start asking, could this be true? ok, it may be heretical (or whatever modern equivalent), but might it also be true?

heresy

this won't get us all the answers, though. what if no one happens to have gotten in trouble for a particular idea yet? what if some idea would be so radioactively controversial that no one would dare express it in public? how can we find these too?

another approach is to follow that word, heresy. in every period of history, there seem to have been labels that got applied to statements to shoot them down before anyone had a chance to ask if they were true or not. "blasphemy", "sacrilege", and "heresy" were such labels for a good part of western history, as in more recent times "indecent", "improper", and "unamerican" have been. by now these labels have lost their sting. they always do. by now they're mostly used ironically. but in their time, they had real force.

the word "defeatist", for example, has no particular political connotations now. but in germany in 1917 it was a weapon, used by ludendorff in a purge of those who favored a negotiated peace. at the start of world war ii it was used extensively by churchill and his supporters to silence their opponents. in 1940, any argument against churchill's aggressive policy was "defeatist". was it right or wrong? ideally, no one got far enough to ask that.

we have such labels today, of course, quite a lot of them, from the all-purpose "inappropriate" to the dreaded "divisive." in any period, it should be easy to figure out what such labels are, simply by looking at what people call ideas they disagree with besides untrue. when a politician says his opponent is mistaken, that's a straightforward criticism, but when he attacks a statement as "divisive" or "racially insensitive" instead of arguing that it's false, we should start paying attention.

so another way to figure out which of our taboos future generations will laugh at is to start with the labels. take a label — "sexist", for example — and try to think of some ideas that would be called that. then for each ask, might this be true?

just start listing ideas at random? yes, because they won't really be random. the ideas that come to mind first will be the most plausible ones. they'll be things you've already noticed but didn't let yourself

think.

in 1989 some clever researchers tracked the eye movements of radiologists as they scanned chest images for signs of lung cancer.[3] they found that even when the radiologists missed a cancerous lesion, their eyes had usually paused at the site of it. part of their brain knew there was something there; it just didn't percolate all the way up into conscious knowledge. i think many interesting heretical thoughts are already mostly formed in our minds. if we turn off our self-censorship temporarily, those will be the first to emerge.

time and space

if we could look into the future it would be obvious which of our taboos they'd laugh at. we can't do that, but we can do something almost as good: we can look into the past. another way to figure out what we're getting wrong is to look at what used to be acceptable and is now unthinkable.

changes between the past and the present sometimes do represent progress. in a field like physics, if we disagree with past generations it's because we're right and they're wrong. but this becomes rapidly less true as you move away from the certainty of the hard sciences. by the time you get to social questions, many changes are just fashion. the age of consent fluctuates like hemlines.

we may imagine that we are a great deal smarter and more virtuous than past generations, but the more history you read, the less likely this seems. people in past times were much like us. not heroes, not barbarians. whatever their ideas were, they were ideas reasonable people could believe.

so here is another source of interesting heresies. diff present ideas against those of various past cultures, and see what you get.[4] some will be shocking by present standards. ok, fine; but which might also be true?

you don't have to look into the past to find big differences. in our own time, different societies have wildly varying ideas of what's ok and what isn't. so you can try diffing other cultures' ideas against ours as well. (the best way to do that is to visit them.)

any idea that's considered harmless in a significant percentage of times and places, and yet is taboo in ours, is a candidate for something we're mistaken about.

for example, at the high water mark of political correctness in the early 1990s, harvard distributed to its faculty and staff a brochure saying, among other things, that it was inappropriate to compliment a colleague or student's clothes. no more "nice shirt." i think this principle is rare among the world's cultures, past or present. there are probably more where it's considered especially polite to compliment someone's clothing than where it's considered improper.

odds are this is, in a mild form, an example of one of the taboos a visitor from the future would have to be careful to avoid if he happened to set his time machine for cambridge, massachusetts, 1992.[5]

prigs

of course, if they have time machines in the future they'll probably have a separate reference manual just for cambridge. this has always been a fussy place, a town of i dotters and t crossers, where you're liable to get both your grammar and your ideas corrected in the same conversation. and that suggests another way to find taboos. look for prigs, and see what's inside their heads.

kids' heads are repositories of all our taboos. it seems fitting to us that kids' ideas should be bright and clean. the picture we give them of the world is not merely simplified, to suit their developing minds, but sanitized as well, to suit our ideas of what kids ought to think.[6]

you can see this on a small scale in the matter of dirty words. a lot of my friends are starting to have children now, and they're all trying not to use words like "fuck" and "shit" within baby's hearing, lest baby start using these words too. but these words are part of the language, and adults use them all the time. so parents are giving their kids an inaccurate idea of the language by not using them. why do they do this? because they don't think it's fitting that kids should use the whole language. we like children to seem innocent.[7]

most adults, likewise, deliberately give kids a misleading view of the world. one of the most obvious examples is santa claus. we think it's cute for little kids to believe in santa claus. i myself think it's cute for little kids to believe in santa claus. but one wonders, do we tell them this stuff for their sake, or for ours?

i'm not arguing for or against this idea here. it is probably inevitable that parents should want to dress up their kids' minds in cute little baby outfits. i'll probably do it myself. the important thing for our purposes is that, as a result, a well brought-up teenage kid's brain is a more or less complete collection of all our taboos — and in mint condition, because they're untainted by experience. whatever we think that will later turn out to be ridiculous, it's almost certainly inside that head.

how do we get at these ideas? by the following thought experiment. imagine a kind of latter-day conrad character who has worked for a time as a mercenary in africa, for a time as a doctor in nepal, for a time as the manager of a nightclub in miami. the specifics don't matter — just someone who has seen a lot. now imagine comparing what's inside this guy's head with what's inside the head of a well-behaved sixteen year old girl from the suburbs. what does he think that would shock her? he knows the world; she knows, or at least embodies, present taboos. subtract one from the other, and the result is what we can't say.

mechanism

i can think of one more way to figure out what we can't say: to look at how taboos are created. how do moral fashions arise, and why are they adopted? if we can understand this mechanism, we may be able to see it at work in our own time.

moral fashions don't seem to be created the way ordinary fashions are. ordinary fashions seem to arise by accident when everyone imitates the whim of some influential person. the fashion for broad-toed shoes in late fifteenth century europe began because charles viii of france had six toes on one foot. the fashion for the name gary began when the actor frank cooper adopted the name of a tough mill town in indiana. moral fashions more often seem to be created deliberately. when there's something we can't say, it's often because some group doesn't want us to.

the prohibition will be strongest when the group is nervous. the irony of galileo's situation was that he got in trouble for repeating copernicus's ideas. copernicus himself didn't. in fact, copernicus was a canon of a cathedral, and dedicated his book to the pope. but by galileo's time the church was in the throes of the counter-reformation and was much more worried about unorthodox ideas.

to launch a taboo, a group has to be poised halfway between weakness and power. a confident group doesn't need taboos to protect it. it's not considered improper to make disparaging remarks about americans, or the english. and yet a group has to be powerful enough to enforce a taboo. coprophiles, as of this writing, don't seem to be numerous or energetic enough to have had their interests promoted to a lifestyle.

i suspect the biggest source of moral taboos will turn out to be power struggles in which one side only barely has the upper hand. that's where you'll find a group powerful enough to enforce taboos, but

weak enough to need them.

most struggles, whatever they're really about, will be cast as struggles between competing ideas. the english reformation was at bottom a struggle for wealth and power, but it ended up being cast as a struggle to preserve the souls of englishmen from the corrupting influence of rome. it's easier to get people to fight for an idea. and whichever side wins, their ideas will also be considered to have triumphed, as if god wanted to signal his agreement by selecting that side as the victor.

we often like to think of world war ii as a triumph of freedom over totalitarianism. we conveniently forget that the soviet union was also one of the winners.

i'm not saying that struggles are never about ideas, just that they will always be made to seem to be about ideas, whether they are or not. and just as there is nothing so unfashionable as the last, discarded fashion, there is nothing so wrong as the principles of the most recently defeated opponent.

representational art is only now recovering from the approval of both hitler and stalin.[8]

although moral fashions tend to arise from different sources than fashions in clothing, the mechanism of their adoption seems much the same. the early adopters will be driven by ambition: self-consciously cool people who want to distinguish themselves from the common herd. as the fashion becomes established they'll be joined by a second, much larger group, driven by fear.[9] this second group adopt the fashion not because they want to stand out but because they are afraid of standing out.

so if you want to figure out what we can't say, look at the machinery of fashion and try to predict what it would make unsayable. what groups are powerful but nervous, and what ideas would they like to suppress? what ideas were tarnished by association when they ended up on the losing side of a recent struggle? if a self-consciously cool person wanted to differentiate himself from preceding fashions (e.g. from his parents), which of their ideas would he tend to reject? what are conventional-minded people afraid of saying?

this technique won't find us all the things we can't say. i can think of some that aren't the result of any recent struggle. many of our taboos are rooted deep in the past. but this approach, combined with the preceding four, will turn up a good number of unthinkable ideas.

why

some would ask, why would one want to do this? why deliberately go poking around among nasty, disreputable ideas? why look under rocks?

i do it, first of all, for the same reason i did look under rocks as a kid: plain curiosity. and i'm especially curious about anything that's forbidden. let me see and decide for myself.

second, i do it because i don't like the idea of being mistaken. if, like other eras, we believe things that will later seem ridiculous, i want to know what they are so that i, at least, can avoid believing them.

third, i do it because it's good for the brain. to do good work you need a brain that can go anywhere. and you especially need a brain that's in the habit of going where it's not supposed to.

great work tends to grow out of ideas that others have overlooked, and no idea is so overlooked as one that's unthinkable. natural selection, for example. it's so simple. why didn't anyone think of it before? well, that is all too obvious. darwin himself was careful to tiptoe around the implications of his theory. he wanted to spend his time thinking about biology, not arguing with people who accused him of being an atheist.

in the sciences, especially, it's a great advantage to be able to question assumptions. the m.o. of scientists, or at least of the good ones, is precisely that: look for places where conventional wisdom is broken, and then try to pry apart the cracks and see what's underneath. that's where new theories come from.

a good scientist, in other words, does not merely ignore conventional wisdom, but makes a special effort to break it. scientists go looking for trouble. this should be the m.o. of any scholar, but scientists seem much more willing to look under rocks.[10]

why? it could be that the scientists are simply smarter; most physicists could, if necessary, make it through a phd program in french literature, but few professors of french literature could make it through a phd program in physics. or it could be because it's clearer in the sciences whether theories are true or false, and this makes scientists bolder. (or it could be that, because it's clearer in the sciences whether theories are true or false, you have to be smart to get jobs as a scientist, rather than just a good politician.)

whatever the reason, there seems a clear correlation between intelligence and willingness to consider shocking ideas. this isn't just because smart people actively work to find holes in conventional thinking. i think conventions also have less hold over them to start with. you can see that in the way they dress.

it's not only in the sciences that heresy pays off. in any competitive field, you can win big by seeing things that others daren't. and in every field there are probably heresies few dare utter. within the us car industry there is a lot of hand-wringing now about declining market share. yet the cause is so obvious that any observant outsider could explain it in a second: they make bad cars. and they have for so long that by now the us car brands are antibrands — something you'd buy a car despite, not because of. cadillac stopped being the cadillac of cars in about 1970. and yet i suspect no one dares say this.[11] otherwise these companies would have tried to fix the problem.

training yourself to think unthinkable thoughts has advantages beyond the thoughts themselves. it's like stretching. when you stretch before running, you put your body into positions much more extreme than any it will assume during the run. if you can think things so outside the box that they'd make people's hair stand on end, you'll have no trouble with the small trips outside the box that people call innovative.

pensieri stretti

when you find something you can't say, what do you do with it? my advice is, don't say it. or at least, pick your battles.

suppose in the future there is a movement to ban the color yellow. proposals to paint anything yellow are denounced as "yellowist", as is anyone suspected of liking the color. people who like orange are tolerated but viewed with suspicion. suppose you realize there is nothing wrong with yellow. if you go around saying this, you'll be denounced as a yellowist too, and you'll find yourself having a lot of arguments with anti-yellowists. if your aim in life is to rehabilitate the color yellow, that may be what you want. but if you're mostly interested in other questions, being labelled as a yellowist will just be a distraction. argue with idiots, and you become an idiot.

the most important thing is to be able to think what you want, not to say what you want. and if you feel you have to say everything you think, it may inhibit you from thinking improper thoughts. i think it's better to follow the opposite policy. draw a sharp line between your thoughts and your speech. inside your head, anything is allowed. within my head i make a point of encouraging the most outrag-

eous thoughts i can imagine. but, as in a secret society, nothing that happens within the building should be told to outsiders. the first rule of fight club is, you do not talk about fight club.

when milton was going to visit italy in the 1630s, sir henry wootton, who had been ambassador to venice, told him his motto should be "*i pensieri stretti & il viso sciolto.*" closed thoughts and an open face. smile at everyone, and don't tell them what you're thinking. this was wise advice. milton was an argumentative fellow, and the inquisition was a bit restive at that time. but i think the difference between milton's situation and ours is only a matter of degree. every era has its heresies, and if you don't get imprisoned for them you will at least get in enough trouble that it becomes a complete distraction.

i admit it seems cowardly to keep quiet. when i read about the harassment to which the scientologists subject their critics[12], or that pro-israel groups are "compiling dossiers" on those who speak out against israeli human rights abuses[13], or about people being sued for violating the dmca[14], part of me wants to say, "all right, you bastards, bring it on." the problem is, there are so many things you can't say. if you said them all you'd have no time left for your real work. you'd have to turn into noam chomsky.[15]

the trouble with keeping your thoughts secret, though, is that you lose the advantages of discussion. talking about an idea leads to more ideas. so the optimal plan, if you can manage it, is to have a few trusted friends you can speak openly to. this is not just a way to develop ideas; it's also a good rule of thumb for choosing friends. the people you can say heretical things to without getting jumped on are also the most interesting to know.

viso sciolto?

i don't think we need the *viso sciolto* so much as the *pensieri stretti.* perhaps the best policy is to make it plain that you don't agree with whatever zealotry is current in your time, but not to be too specific about what you disagree with. zealots will try to draw you out, but you don't have to answer them. if they try to force you to treat a question on their terms by asking "are you with us or against us?" you can always just answer "neither".

better still, answer "i haven't decided." that's what larry summers did when a group tried to put him in this position. explaining himself later, he said "i don't do litmus tests."[16] a lot of the questions people get hot about are actually quite complicated. there is no prize for getting the answer quickly.

if the anti-yellowists seem to be getting out of hand and you want to fight back, there are ways to do it without getting yourself accused of being a yellowist. like skirmishers in an ancient army, you want to avoid directly engaging the main body of the enemy's troops. better to harass them with arrows from a distance.

one way to do this is to ratchet the debate up one level of abstraction. if you argue against censorship in general, you can avoid being accused of whatever heresy is contained in the book or film that someone is trying to censor. you can attack labels with meta-labels: labels that refer to the use of labels to prevent discussion. the spread of the term "political correctness" meant the beginning of the end of political correctness, because it enabled one to attack the phenomenon as a whole without being accused of any of the specific heresies it sought to suppress.

another way to counterattack is with metaphor. arthur miller undermined the house un-american activities committee by writing a play, "the crucible," about the salem witch trials. he never referred directly to the committee and so gave them no way to reply. what could huac do, defend the salem witch trials? and yet miller's metaphor stuck so well that to this day the activities of the committee are often described as a "witch-hunt."

best of all, probably, is humor. zealots, whatever their cause, invariably lack a sense of humor. they can't reply in kind to jokes. they're as unhappy on the territory of humor as a mounted knight on a skating rink. victorian prudishness, for example, seems to have been defeated mainly by treating it as a joke. likewise its reincarnation as political correctness. "i am glad that i managed to write 'the crucible,'" arthur miller wrote, "but looking back i have often wished i'd had the temperament to do an absurd comedy, which is what the situation deserved."[17]

abq

a dutch friend says i should use holland as an example of a tolerant society. it's true they have a long tradition of comparative open-mindedness. for centuries the low countries were the place to go to say things you couldn't say anywhere else, and this helped to make the region a center of scholarship and industry (which have been closely tied for longer than most people realize). descartes, though claimed by the french, did much of his thinking in holland.

and yet, i wonder. the dutch seem to live their lives up to their necks in rules and regulations. there's so much you can't do there; is there really nothing you can't say?

certainly the fact that they value open-mindedness is no guarantee. who thinks they're not open-minded? our hypothetical prim miss from the suburbs thinks she's open-minded. hasn't she been taught to be? ask anyone, and they'll say the same thing: they're pretty open-minded, though they draw the line at things that are really wrong. (some tribes may avoid "wrong" as judgemental, and may instead use a more neutral sounding euphemism like "negative" or "destructive".)

when people are bad at math, they know it, because they get the wrong answers on tests. but when people are bad at open-mindedness they don't know it. in fact they tend to think the opposite. remember, it's the nature of fashion to be invisible. it wouldn't work otherwise. fashion doesn't seem like fashion to someone in the grip of it. it just seems like the right thing to do. it's only by looking from a distance that we see oscillations in people's idea of the right thing to do, and can identify them as fashions.

time gives us such distance for free. indeed, the arrival of new fashions makes old fashions easy to see, because they seem so ridiculous by contrast. from one end of a pendulum's swing, the other end seems especially far away.

to see fashion in your own time, though, requires a conscious effort. without time to give you distance, you have to create distance yourself. instead of being part of the mob, stand as far away from it as you can and watch what it's doing. and pay especially close attention whenever an idea is being suppressed. web filters for children and employees often ban sites containing pornography, violence, and hate speech. what counts as pornography and violence? and what, exactly, is "hate speech?" this sounds like a phrase out of *1984*.

labels like that are probably the biggest external clue. if a statement is false, that's the worst thing you can say about it. you don't need to say that it's heretical. and if it isn't false, it shouldn't be suppressed. so when you see statements being attacked as x-ist or y-ic (substitute your current values of x and y), whether in 1630 or 2030, that's a sure sign that something is wrong. when you hear such labels being used, ask why.

especially if you hear yourself using them. it's not just the mob you need to learn to watch from a distance. you need to be able to watch your own thoughts from a distance. that's not a radical idea, by the way; it's the main difference between children and adults. when a child gets angry because he's tired, he doesn't know what's happening. an adult can distance himself enough from the situation to say

"never mind, i'm just tired." i don't see why one couldn't, by a similar process, learn to recognize and discount the effects of moral fashions.

you have to take that extra step if you want to think clearly. but it's harder, because now you're working against social customs instead of with them. everyone encourages you to grow up to the point where you can discount your own bad moods. few encourage you to continue to the point where you can discount society's bad moods.

how can you see the wave, when you're the water? always be questioning. that's the only defence. what can't you say? and why?

---

notes

thanks to sarah harlin, trevor blackwell, jessica livingston, robert morris, eric raymond and bob van der zwaan for reading drafts of this essay, and to lisa randall, jackie mcdonough, ryan stanley and joel rainey for conversations about heresy. needless to say they bear no blame for opinions expressed in it, and especially for opinions *not* expressed in it.

02

inequality and risk

august 2005
paulgraham.com/inequality.html

(this essay is derived from a talk at defcon 2005.)

suppose you wanted to get rid of economic inequality. there are two ways to do it: give money to the poor, or take it away from the rich. but they amount to the same thing, because if you want to give money to the poor, you have to get it from somewhere. you can't get it from the poor, or they just end up where they started. you have to get it from the rich.

there is of course a way to make the poor richer without simply shifting money from the rich. you could help the poor become more productive — for example, by improving access to education. instead of taking money from engineers and giving it to checkout clerks, you could enable people who would have become checkout clerks to become engineers.

this is an excellent strategy for making the poor richer. but the evidence of the last 200 years shows that it doesn't reduce economic inequality, because it makes the rich richer too. if there are more engineers, then there are more opportunities to hire them and to sell them things. henry ford couldn't have made a fortune building cars in a society in which most people were still subsistence farmers; he would have had neither workers nor customers.

if you want to reduce economic inequality instead of just improving the overall standard of living, it's not enough just to raise up the poor. what if one of your newly minted engineers gets ambitious and goes on to become another bill gates? economic inequality will be as bad as ever. if you actually want to compress the gap between rich and poor, you have to push down on the top as well as pushing up on the bottom.

how do you push down on the top? you could try to decrease the productivity of the people who make the most money: make the best surgeons operate with their left hands, force popular actors to overeat, and so on. but this approach is hard to implement. the only practical solution is to let people do the best work they can, and then (either by taxation or by limiting what they can charge) to confiscate whatever you deem to be surplus.

so let's be clear what reducing economic inequality means. it is identical with taking money from the rich.

when you transform a mathematical expression into another form, you often notice new things. so it is in this case. taking money from the rich turns out to have consequences one might not foresee when one phrases the same idea in terms of "reducing inequality."

the problem is, risk and reward have to be proportionate. a bet with only a 10% chance of winning has to pay more than one with a 50% chance of winning, or no one will take it. so if you lop off the top of the possible rewards, you thereby decrease people's willingness to take risks.

transposing into our original expression, we get: decreasing economic inequality means decreasing the risk people are willing to take.

there are whole classes of risks that are no longer worth taking if the maximum return is decreased. one reason high tax rates are disastrous is that this class of risks includes starting new companies.

investors

startups are intrinsically risky. a startup is like a small boat in the open sea. one big wave and you're sunk. a competing product, a downturn in the economy, a delay in getting funding or regulatory approval, a patent suit, changing technical standards, the departure of a key employee, the loss of a big account — any one of these can destroy you overnight. it seems only about 1 in 10 startups succeeds.[1]

our startup paid its first round of outside investors 36x. which meant, with current us tax rates, that it made sense to invest in us if we had better than a 1 in 24 chance of succeeding. that sounds about right. that's probably roughly how we looked when we were a couple of nerds with no business experience operating out of an apartment.

if that kind of risk doesn't pay, venture investing, as we know it, doesn't happen.

that might be ok if there were other sources of capital for new companies. why not just have the government, or some large almost-government organization like fannie mae, do the venture investing instead of private funds?

i'll tell you why that wouldn't work. because then you're asking government or almost-government employees to do the one thing they are least able to do: take risks.

as anyone who has worked for the government knows, the important thing is not to make the right choices, but to make choices that can be justified later if they fail. if there is a safe option, that's the one a bureaucrat will choose. but that is exactly the wrong way to do venture investing. the nature of the business means that you want to make terribly risky choices, if the upside looks good enough.

vcs are currently paid in a way that makes them focus on the upside: they get a percentage of the fund's gains. and that helps overcome their understandable fear of investing in a company run by nerds who look like (and perhaps are) college students.

if vcs weren't allowed to get rich, they'd behave like bureaucrats. without hope of gain, they'd have only fear of loss. and so they'd make the wrong choices. they'd turn down the nerds in favor of the smooth-talking mba in a suit, because that investment would be easier to justify later if it failed.

founders

but even if you could somehow redesign venture funding to work without allowing vcs to become rich, there's another kind of investor you simply cannot replace: the startups' founders and early employees.

what they invest is their time and ideas. but these are equivalent to money; the proof is that investors are willing (if forced) to treat them as interchangeable, granting the same status to "sweat equity" and the equity they've purchased with cash.

the fact that you're investing time doesn't change the relationship between risk and reward. if you're going to invest your time in something with a small chance of succeeding, you'll only do it if there is a proportionately large payoff.[2] if large payoffs aren't allowed, you may as well play it safe.

like many startup founders, i did it to get rich. but not because i wanted to buy expensive things. what i wanted was security. i wanted to make enough money that i didn't have to worry about money. if i'd been forbidden to make enough from a startup to do this, i would have sought security by some other means: for example, by going to work for a big, stable organization from which it would be hard to get fired. instead of busting my ass in a startup, i would have tried to get a nice, low-stress job at a big research lab, or tenure at a university.

that's what everyone does in societies where risk isn't rewarded. if you can't ensure your own security, the next best thing is to make a nest for yourself in some large organization where your status depends mostly on seniority.[3]

even if we could somehow replace investors, i don't see how we could replace founders. investors mainly contribute money, which in principle is the same no matter what the source. but the founders contribute ideas. you can't replace those.

let's rehearse the chain of argument so far. i'm heading for a conclusion to which many readers will have to be dragged kicking and screaming, so i've tried to make each link unbreakable. decreasing economic inequality means taking money from the rich. since risk and reward are equivalent, decreasing potential rewards automatically decreases people's appetite for risk. startups are intrinsically risky. without the prospect of rewards proportionate to the risk, founders will not invest their time in a startup. founders are irreplaceable. so eliminating economic inequality means eliminating startups.

economic inequality is not just a consequence of startups. it's the engine that drives them, in the same way a fall of water drives a water mill. people start startups in the hope of becoming much richer than they were before. and if your society tries to prevent anyone from being much richer than anyone else, it will also prevent one person from being much richer at t2 than t1.

growth

this argument applies proportionately. it's not just that if you eliminate economic inequality, you get no startups. to the extent you reduce economic inequality, you decrease the number of startups.[4] increase taxes, and willingness to take risks decreases in proportion.

and that seems bad for everyone. new technology and new jobs both come disproportionately from new companies. indeed, if you don't have startups, pretty soon you won't have established companies either, just as, if you stop having kids, pretty soon you won't have any adults.

it sounds benevolent to say we ought to reduce economic inequality. when you phrase it that way, who can argue with you? *inequality* has to be bad, right? it sounds a good deal less benevolent to say we ought to reduce the rate at which new companies are founded. and yet the one implies the other.

indeed, it may be that reducing investors' appetite for risk doesn't merely kill off larval startups, but kills off the most promising ones especially. startups yield faster growth at greater risk than established companies. does this trend also hold among startups? that is, are the riskiest startups the ones that generate most growth if they succeed? i suspect the answer is yes. and that's a chilling thought, because it means that if you cut investors' appetite for risk, the most beneficial startups are the first to go.

not all rich people got that way from startups, of course. what if we let people get rich by starting startups, but taxed away all other surplus wealth? wouldn't that at least decrease inequality?

less than you might think. if you made it so that people could only get rich by starting startups, people who wanted to get rich would all start startups. and that might be a great thing. but i don't think it would have much effect on the distribution of wealth. people who want to get rich will do whatever they have to. if startups are the only way to do it, you'll just get far more people starting startups. (if you write the laws very carefully, that is. more likely, you'll just get a lot of people doing things that can be made to look on paper like startups.)

if we're determined to eliminate economic inequality, there is still one way out: we could say that we're willing to go ahead and do without startups. what would happen if we did?

at a minimum, we'd have to accept lower rates of technological growth. if you believe that large, established companies could somehow be made to develop new technology as fast as startups, the ball is in your court to explain how. (if you can come up with a remotely plausible story, you can make a fortune writing business books and consulting for large companies.)[5]

ok, so we get slower growth. is that so bad? well, one reason it's bad in practice is that other countries might not agree to slow down with us. if you're content to develop new technologies at a slower rate than the rest of the world, what happens is that you don't invent anything at all. anything you might

discover has already been invented elsewhere. and the only thing you can offer in return is raw materials and cheap labor. once you sink that low, other countries can do whatever they like with you: install puppet governments, siphon off your best workers, use your women as prostitutes, dump their toxic waste on your territory — all the things we do to poor countries now. the only defense is to isolate yourself, as communist countries did in the twentieth century. but the problem then is, you have to become a police state to enforce it.

wealth and power

i realize startups are not the main target of those who want to eliminate economic inequality. what they really dislike is the sort of wealth that becomes self-perpetuating through an alliance with power. for example, construction firms that fund politicians' campaigns in return for government contracts, or rich parents who get their children into good colleges by sending them to expensive schools designed for that purpose. but if you try to attack this type of wealth through *economic* policy, it's hard to hit without destroying startups as collateral damage.

the problem here is not wealth, but corruption. so why not go after corruption?

we don't need to prevent people from being rich if we can prevent wealth from translating into power. and there has been progress on that front. before he died of drink in 1925, commodore vanderbilt's wastrel grandson reggie ran down pedestrians on five separate occasions, killing two of them. by 1969, when ted kennedy drove off the bridge at chappaquiddick, the limit seemed to be down to one. today it may well be zero. but what's changed is not variation in wealth. what's changed is the ability to translate wealth into power.

how do you break the connection between wealth and power? demand transparency. watch closely how power is exercised, and demand an account of how decisions are made. why aren't all police interrogations videotaped? why did 36% of princeton's class of 2007 come from prep schools, when only 1.7% of american kids attend them? why did the us really invade iraq? why don't government officials disclose more about their finances, and why only during their term of office?

a friend of mine who knows a lot about computer security says the single most important step is to log everything. back when he was a kid trying to break into computers, what worried him most was the idea of leaving a trail. he was more inconvenienced by the need to avoid that than by any obstacle deliberately put in his path.

like all illicit connections, the connection between wealth and power flourishes in secret. expose all transactions, and you will greatly reduce it. log everything. that's a strategy that already seems to be working, and it doesn't have the side effect of making your whole country poor.

i don't think many people realize there is a connection between economic inequality and risk. i didn't fully grasp it till recently. i'd known for years of course that if one didn't score in a startup, the other alternative was to get a cozy, tenured research job. but i didn't understand the equation governing my behavior. likewise, it's obvious empirically that a country that doesn't let people get rich is headed for disaster, whether it's diocletian's rome or harold wilson's britain. but i did not till recently understand the role risk played.

if you try to attack wealth, you end up nailing risk as well, and with it growth. if we want a fairer world, i think we're better off attacking one step downstream, where wealth turns into power.

notes

1 success here is defined from the initial investors' point of view: either an ipo, or an acquisition for more than the valuation at the last round of funding. the conventional 1 in 10 success rate is suspiciously neat, but conversations with vcs suggest it's roughly correct for startups overall. top vc firms expect to do better.

2 i'm not claiming founders sit down and calculate the expected after-tax return from a startup. they're motivated by examples of other people who did it. and those examples do reflect after-tax returns.

3 conjecture: the variation in wealth in a (non-corrupt) country or organization will be inversely proportional to the prevalence of systems of seniority. so if you suppress variation in wealth, seniority will become correspondingly more important. so far, i know of no counterexamples, though in very corrupt countries you may get both simultaneously. (thanks to daniel sobral for pointing this out.)

4 in a country with a truly feudal economy, you might be able to redistribute wealth successfully, because there are no startups to kill.

5 the speed at which startups develop new techology is the other reason they pay so well. as i explained in "how to make wealth", what you do in a startup is compress a lifetime's worth of work into a few years. it seems as dumb to discourage that as to discourage risk-taking.

thanks to chris anderson, trevor blackwell, dan giffin, jessica livingston, and evan williams for reading drafts of this essay, and to langley steinert, sangam pant, and mike moritz for information about venture investing.

03

economic inequality

january 2016
paulgraham.com/ineq.html

since the 1970s, economic inequality in the us has increased dramatically. and in particular, the rich have gotten a lot richer. nearly everyone who writes about the topic says that economic inequality should be decreased.

i'm interested in this question because i was one of the founders of a company called y combinator that helps people start startups. almost by definition, if a startup succeeds, its founders become rich. which means by helping startup founders i've been helping to increase economic inequality. if economic inequality should be decreased, i shouldn't be helping founders. no one should be.

but that doesn't sound right. what's going on here? what's going on is that while economic inequality is a single measure (or more precisely, two: variation in income, and variation in wealth), it has multiple causes. many of these causes are bad, like tax loopholes and drug addiction. but some are good, like larry page and sergey brin starting the company you use to find things online.

if you want to understand economic inequality — and more importantly, if you actually want to fix the bad aspects of it — you have to tease apart the components. and yet the trend in nearly everything written about the subject is to do the opposite: to squash together all the aspects of economic inequality as if it were a single phenomenon.

sometimes this is done for ideological reasons. sometimes it's because the writer only has very high-level data and so draws conclusions from that, like the proverbial drunk who looks for his keys under the lamppost, instead of where he dropped them, because the light is better there. sometimes it's because the writer doesn't understand critical aspects of inequality, like the role of technology in wealth creation. much of the time, perhaps most of the time, writing about economic inequality combines all three.

—

the most common mistake people make about economic inequality is to treat it as a single phenomenon. the most naive version of which is the one based on the pie fallacy: that the rich get rich by taking money from the poor.

usually this is an assumption people start from rather than a conclusion they arrive at by examining the evidence. sometimes the pie fallacy is stated explicitly:

> ...those at the top are grabbing an increasing fraction of the nation's income — so much of a larger share that what's left over for the rest is diminished.... [1]

other times it's more unconscious. but the unconscious form is very widespread. i think because we grow up in a world where the pie fallacy is actually true. to kids, wealth *is* a fixed pie that's shared out, and if one person gets more, it's at the expense of another. it takes a conscious effort to remind oneself that the real world doesn't work that way.

in the real world you can create wealth as well as taking it from others. a woodworker creates wealth. he makes a chair, and you willingly give him money in return for it. a high-frequency trader does not. he makes a dollar only when someone on the other end of a trade loses a dollar.

if the rich people in a society got that way by taking wealth from the poor, then you have the degenerate case of economic inequality, where the cause of poverty is the same as the cause of wealth. but instances of inequality don't have to be instances of the degenerate case. if one woodworker makes 5 chairs and another makes none, the second woodworker will have less money, but not because anyone took anything from him.

even people sophisticated enough to know about the pie fallacy are led toward it by the custom of describing economic inequality as a ratio of one quantile's income or wealth to another's. it's so easy to slip from talking about income shifting from one quantile to another, as a figure of speech, into believing that is literally what's happening.

except in the degenerate case, economic inequality can't be described by a ratio or even a curve. in the general case it consists of multiple ways people become poor, and multiple ways people become rich. which means to understand economic inequality in a country, you have to go find individual people who are poor or rich and figure out why. [2]

if you want to understand *change* in economic inequality, you should ask what those people would have done when it was different. this is one way i know the rich aren't all getting richer simply from some new system for transferring wealth to them from everyone else. when you use the would-have method with startup founders, you find what most would have done back in 1960, when economic inequality was lower, was to join big companies or become professors. before mark zuckerberg started facebook, his default expectation was that he'd end up working at microsoft. the reason he and most other startup founders are richer than they would have been in the mid 20th century is not because of some right turn the country took during the reagan administration, but because progress in technology has made it much easier to start a new company that grows fast.

traditional economists seem strangely averse to studying individual humans. it seems to be a rule with them that everything has to start with statistics. so they give you very precise numbers about variation in wealth and income, then follow it with the most naive speculation about the underlying causes.

but while there are a lot of people who get rich through rent-seeking of various forms, and a lot who get rich by playing zero-sum games, there are also a significant number who get rich by creating wealth. and creating wealth, as a source of economic inequality, is different from taking it — not just morally, but also practically, in the sense that it is harder to eradicate. one reason is that variation in productivity is accelerating. the rate at which individuals can create wealth depends on the technology available to them, and that grows exponentially. the other reason creating wealth is such a tenacious source of inequality is that it can expand to accommodate a lot of people.

—

i'm all for shutting down the crooked ways to get rich. but that won't eliminate great variations in wealth, because as long as you leave open the option of getting rich by creating wealth, people who want to get rich will do that instead.

most people who get rich tend to be fairly driven. whatever their other flaws, laziness is usually not one of them. suppose new policies make it hard to make a fortune in finance. does it seem plausible that the people who currently go into finance to make their fortunes will continue to do so, but be content to work for ordinary salaries? the reason they go into finance is not because they love finance but because they want to get rich. if the only way left to get rich is to start startups, they'll start startups. they'll do well at it too, because determination is the main factor in the success of a startup. [3] and while it would probably be a good thing for the world if people who wanted to get rich switched from playing zero-sum games to creating wealth, that would not only not eliminate great variations in wealth, but might even exacerbate them. in a zero-sum game there is at least a limit to the upside. plus a lot of the new startups would create new technology that further accelerated variation in productivity.

variation in productivity is far from the only source of economic inequality, but it is the irreducible core of it, in the sense that you'll have that left when you eliminate all other sources. and if you do, that core will be big, because it will have expanded to include the efforts of all the refugees. plus it will have a large baumol penumbra around it: anyone who could get rich by creating wealth on their own account will have to be paid enough to prevent them from doing it.

you can't prevent great variations in wealth without preventing people from getting rich, and you can't do that without preventing them from starting startups.

so let's be clear about that. eliminating great variations in wealth would mean eliminating startups. and that doesn't seem a wise move. especially since it would only mean you eliminated startups in your own country. ambitious people already move halfway around the world to further their careers, and startups can operate from anywhere nowadays. so if you made it impossible to get rich by creating wealth in your country, people who wanted to do that would just leave and do it somewhere else. which would certainly get you a lower gini coefficient, along with a lesson in being careful what you ask for. [4]

i think rising economic inequality is the inevitable fate of countries that don't choose something worse. we had a 40 year stretch in the middle of the 20th century that convinced some people otherwise. but as i explained in the refragmentation, that was an anomaly — a unique combination of circumstances that compressed american society not just economically but culturally too. [5]

and while some of the growth in economic inequality we've seen since then has been due to bad behavior of various kinds, there has simultaneously been a huge increase in individuals' ability to create wealth. startups are almost entirely a product of this period. and even within the startup world, there has been a qualitative change in the last 10 years. technology has decreased the cost of starting a startup so much that founders now have the upper hand over investors. founders get less diluted, and it is now common for them to retain board control as well. both further increase economic inequality, the former because founders own more stock, and the latter because, as investors have learned, founders tend to be better at running their companies than investors.

while the surface manifestations change, the underlying forces are very, very old. the acceleration of productivity we see in silicon valley has been happening for thousands of years. if you look at the history of stone tools, technology was already accelerating in the mesolithic. the acceleration would have been too slow to perceive in one lifetime. such is the nature of the leftmost part of an exponential curve. but it was the same curve.

you do not want to design your society in a way that's incompatible with this curve. the evolution of technology is one of the most powerful forces in history.

louis brandeis said "we may have democracy, or we may have wealth concentrated in the hands of a few, but we can't have both." that sounds plausible. but if i have to choose between ignoring him and ignoring an exponential curve that has been operating for thousands of years, i'll bet on the curve. ignoring any trend that has been operating for thousands of years is dangerous. but exponential growth, especially, tends to bite you.

—

if accelerating variation in productivity is always going to produce some baseline growth in economic inequality, it would be a good idea to spend some time thinking about that future. can you have a healthy society with great variation in wealth? what would it look like?

notice how novel it feels to think about that. the public conversation so far has been exclusively about the need to decrease economic inequality. we've barely given a thought to how to live with it.

i'm hopeful we'll be able to. brandeis was a product of the gilded age, and things have changed since then. it's harder to hide wrongdoing now. and to get rich now you don't have to buy politicians the way railroad or oil magnates did. [6] the great concentrations of wealth i see around me in silicon valley don't seem to be destroying democracy.

there are lots of things wrong with the us that have economic inequality as a symptom. we should fix those things. in the process we may decrease economic inequality. but we can't start from the symptom and hope to fix the underlying causes. [7]

the most obvious is poverty. i'm sure most of those who want to decrease economic inequality want to do it mainly to help the poor, not to hurt the rich. [8] indeed, a good number are merely being sloppy by speaking of decreasing economic inequality when what they mean is decreasing poverty. but this is a situation where it would be good to be precise about what we want. poverty and economic inequality are not identical. when the city is turning off your water because you can't pay the bill, it doesn't make any difference what larry page's net worth is compared to yours. he might only be a few times richer than you, and it would still be just as much of a problem that your water was getting turned off.

closely related to poverty is lack of social mobility. i've seen this myself: you don't have to grow up rich or even upper middle class to get rich as a startup founder, but few successful founders grew up desperately poor. but again, the problem here is not simply economic inequality. there is an enormous difference in wealth between the household larry page grew up in and that of a successful startup founder, but that didn't prevent him from joining their ranks. it's not economic inequality per se that's blocking social mobility, but some specific combination of things that go wrong when kids grow up sufficiently poor.

one of the most important principles in silicon valley is that "you make what you measure." it means that if you pick some number to focus on, it will tend to improve, but that you have to choose the right number, because only the one you choose will improve; another that seems conceptually adjacent might not. for example, if you're a university president and you decide to focus on graduation rates, then you'll improve graduation rates. but only graduation rates, not how much students learn. students could learn less, if to improve graduation rates you made classes easier.

economic inequality is sufficiently far from identical with the various problems that have it as a symptom that we'll probably only hit whichever of the two we aim at. if we aim at economic inequality, we won't fix these problems. so i say let's aim at the problems.

for example, let's attack poverty, and if necessary damage wealth in the process. that's much more likely to work than attacking wealth in the hope that you will thereby fix poverty. [9] and if there are people getting rich by tricking consumers or lobbying the government for anti-competitive regulations or tax loopholes, then let's stop them. not because it's causing economic inequality, but because it's stealing. [10]

if all you have is statistics, it seems like that's what you need to fix. but behind a broad statistical measure like economic inequality there are some things that are good and some that are bad, some that are historical trends with immense momentum and others that are random accidents. if we want to fix the world behind the statistics, we have to understand it, and focus our efforts where they'll do the most good.

notes

1 stiglitz, joseph. *the price of inequality*. norton, 2012. p. 32.

2 particularly since economic inequality is a matter of outliers, and outliers are disproportionately likely to have gotten where they are by ways that have little do with the sort of things economists usually think about, like wages and productivity, but rather by, say, ending up on the wrong side of the "war on drugs."

3 determination is the most important factor in deciding between success and failure, which in startups tend to be sharply differentiated. but it takes more than determination to create one of the hugely successful startups. though most founders start out excited about the idea of getting rich, purely mercenary founders will usually take one of the big acquisition offers most successful startups get on the way up. the founders who go on to the next stage tend to be driven by a sense of mission. they have the same attachment to their companies that an artist or writer has to their work. but it is very hard to predict at the outset which founders will do that. it's not simply a function of their initial attitude. starting a company changes people.

4 after reading a draft of this essay, richard florida told me how he had once talked to a group of europeans "who said they wanted to make europe more entrepreneurial and more like silicon valley. i said by definition this will give you more inequality. they thought i was insane — they could not process it."

5 economic inequality has been decreasing globally. but this is mainly due to the erosion of the kleptocracies that formerly dominated all the poorer countries. once the playing field is leveler politically, we'll see economic inequality start to rise again. the us is the bellwether. the situation we face here, the rest of the world will sooner or later.

6 some people still get rich by buying politicians. my point is that it's no longer a precondition.

7 as well as problems that have economic inequality as a symptom, there are those that have it as a cause. but in most if not all, economic inequality is not the primary cause. there is usually some injustice that is allowing economic inequality to turn into other forms of inequality, and that injustice is what we need to fix. for example, the police in the us treat the poor worse than the rich. but the solution is not to make people richer. it's to make the police treat people more equitably. otherwise they'll continue to maltreat people who are weak in other ways.

8 some who read this essay will say that i'm clueless or even being deliberately misleading by focusing so much on the richer end of economic inequality — that economic inequality is really about poverty. but that is exactly the point i'm making, though sloppier language than i'd use to make it. the real problem is poverty, not economic inequality. and if you conflate them you're aiming at the wrong target.

others will say i'm clueless or being misleading by focusing on people who get rich by creating wealth — that startups aren't the problem, but corrupt practices in finance, healthcare, and so on. once again, that is exactly my point. the problem is not economic inequality, but those specific abuses.

it's a strange task to write an essay about why something isn't the problem, but that's the situation you find yourself in when so many people mistakenly think it is.

9 particularly since many causes of poverty are only partially driven by people trying to make money from them. for example, america's abnormally high incarceration rate is a major cause of poverty. but although for-profit prison companies and prison guard unions both spend a lot lobbying for harsh sentencing laws, they are not the original source of them.

10 incidentally, tax loopholes are definitely not a product of some power shift due to recent increases in economic inequality. the golden age of economic equality in the mid 20th century was also the golden age of tax avoidance. indeed, it was so widespread and so effective that i'm skeptical whether economic inequality was really so low then as we think. in a period when people are trying to hide wealth from the government, it will tend to be hidden from statistics too. one sign of the potential magnitude of the problem is the discrepancy

between government receipts as a percentage of gdp, which have remained more or less constant during the entire period from the end of world war ii to the present, and tax rates, which have varied dramatically.

thanks to sam altman, tiffani ashley bell, patrick collison, ron conway, richard florida, ben horowitz, jessica livingston, robert morris, tim o'reilly, max roser, and alexia tsotsis for reading drafts of this.

note: this is a new version from which i removed a pair of metaphors that made a lot of people mad, essentially by macroexpanding them. if anyone wants to see the old version, i put it here.

related:

04

modeling a wealth tax

august 2020
paulgraham.com/wtax.html

some politicians are proposing to introduce wealth taxes in addition to income and capital gains taxes. let's try modeling the effects of various levels of wealth tax to see what they would mean in practice for a startup founder.

suppose you start a successful startup in your twenties, and then live for another 60 years. how much of your stock will a wealth tax consume?

if the wealth tax applies to all your assets, it's easy to calculate its effect. a wealth tax of 1% means you get to keep 99% of your stock each year. after 60 years the proportion of stock you'll have left will be .99 ^ 60, or .547. so a straight 1% wealth tax means the government will over the course of your life take 45% of your stock.

(losing shares does not, obviously, mean becoming *net* poorer unless the value per share is increasing by less than the wealth tax rate.)

here's how much stock the government would take over 60 years at various levels of wealth tax:

wealth taxgovernment takes 0.1%6%0.5%26% 1.0%45% 2.0%70% 3.0%84% 4.0%91%5.0%95%

a wealth tax will usually have a threshold at which it starts. how much difference would a high threshold make? to model that, we need to make some assumptions about the initial value of your stock and the growth rate.

suppose your stock is initially worth $2 million, and the company's trajectory is as follows: the value of your stock grows 3x for 2 years, then 2x for 2 years, then 50% for 2 years, after which you just get a typical public company growth rate, which we'll call 8%. [1] suppose the wealth tax threshold is $50 million. how much stock does the government take now?

wealth taxgovernment takes 0.1%5%0.5%23% 1.0%41% 2.0%65% 3.0%79% 4.0%88%5.0%93%

it may at first seem surprising that such apparently small tax rates produce such dramatic effects. a 2% wealth tax with a $50 million threshold takes about two thirds of a successful founder's stock.

the reason wealth taxes have such dramatic effects is that they're applied over and over to the same money. income tax happens every year, but only to that year's income. whereas if you live for 60 years after acquiring some asset, a wealth tax will tax that same asset 60 times. a wealth tax compounds.

---

notes

1 in practice, eventually some of this 8% would come in the form of dividends, which are taxed as income at issue, so this model actually represents the most optimistic case for the founder.

05

how to convert between wealth and income tax

may 2026
paulgraham.com/winc.html

how do you convert between wealth and income tax? if a government imposes a wealth tax of 1%, what's the equivalent in income tax?

it's clear from the way most politicians talk about the subject that they not only don't know the answer, but don't even realize there's such a question.

in fact the conversion rate between them is about 20. a wealth tax of 1% is equivalent to an income tax of 20%.

to convert between wealth and income tax rates, you have to divide by the rate of return on capital. the conversion rate of 20 comes from assuming that the risk-free rate of return is 5%. historically that's an optimistic assumption. 4% might be more realistic. but 5% will do. [1]

if we run through an example it will be clear how this works. suppose you have $100, you're getting a 5% rate of return on this capital, and there's a 20% income tax. the 5% rate of return means at the end of one year your $100 has made you another $5. but you have to pay 20% of that, or $1, in income tax, so your after-tax income is $4. at the end of the year, after paying taxes, you have $100 + $4 = $104.

now suppose instead of a 20% income tax, there's a 1% wealth tax. at the end of the year your $100 has made you another $5, as before. but that year you had to pay 1% of your $100, or $1, in wealth tax. so at the end of the year you have $99 + $5 = $104.

each 1% of wealth tax is equivalent to 20% of income tax.

it's clear that politicians don't get this from the way they talk about a "mere 1%" wealth tax. none of them would speak of adding a "mere 20%" to the income tax rate, even though that's mathematically the same thing. [2]

politicians understand that an additional 20% income tax would be a lot. and indeed a us state that added 20% to its top income tax rate would have extraordinarily high taxes.

currently the country with the highest marginal income tax rate is denmark, at 60.5%. the top us federal tax rate is 37%, and the median state income tax rate is oklahoma's, which is 4.75%. so in the median case, a state adding an additional 20% in income tax would have a total marginal tax rate of 37% + 4.75% + 20%, or 61.75%. [3]

in the median case, us state politicians talking about adding a "mere 1%" wealth tax are talking about causing the residents of their state to have the highest taxes in the world. that's not the sort of decision you make lightly.

that's why i think few politicians currently understand how to convert between wealth and income taxes. you can tell from the way they talk about the subject that they don't understand the momentousness of what they're proposing. but i'm optimistic that we can teach them. the answer's not hard to understand, once you realize the question exists.

---

notes

1 it's possible to get a higher rate of return if you're willing to risk losing your capital. but to convert between tax rates you should use the risk-free rate of return, because considered as an anti-investment, a wealth tax is absolutely risk-free: you will absolutely owe the government that money. and while you do have to put "risk-

free" in scare quotes when talking about returns, the kind of risks you're talking about now are the almost apocalyptic kind that would make tax rates a moot point.

2 the same conversion rate applies to capital gains. the source of the multiple is whether the money is taxed every year or just once. indeed it's the same math you'd use to calculate the value of any income-generating asset.

3 you can deduct some state tax from your federal income taxes, but there's a cap on how much you can deduct, which means in the marginal case we simply add the two rates.

thanks to trevor blackwell, jessica livingston, carolynn levy, jon levy, alex tabarrok, and harj taggar for reading drafts of this.

06

the refragmentation

january 2016
paulgraham.com/re.html

one advantage of being old is that you can see change happen in your lifetime. a lot of the change i've seen is fragmentation. us politics is much more polarized than it used to be. culturally we have ever less common ground. the creative class flocks to a handful of happy cities, abandoning the rest. and increasing economic inequality means the spread between rich and poor is growing too. i'd like to propose a hypothesis: that all these trends are instances of the same phenomenon. and moreover, that the cause is not some force that's pulling us apart, but rather the erosion of forces that had been pushing us together.

worse still, for those who worry about these trends, the forces that were pushing us together were an anomaly, a one-time combination of circumstances that's unlikely to be repeated — and indeed, that we would not want to repeat.

the two forces were war (above all world war ii), and the rise of large corporations.

the effects of world war ii were both economic and social. economically, it decreased variation in income. like all modern armed forces, america's were socialist economically. from each according to his ability, to each according to his need. more or less. higher ranking members of the military got more (as higher ranking members of socialist societies always do), but what they got was fixed according to their rank. and the flattening effect wasn't limited to those under arms, because the us economy was conscripted too. between 1942 and 1945 all wages were set by the national war labor board. like the military, they defaulted to flatness. and this national standardization of wages was so pervasive that its effects could still be seen years after the war ended. [1]

business owners weren't supposed to be making money either. fdr said "not a single war millionaire" would be permitted. to ensure that, any increase in a company's profits over prewar levels was taxed at 85%. and when what was left after corporate taxes reached individuals, it was taxed again at a marginal rate of 93%. [2]

socially too the war tended to decrease variation. over 16 million men and women from all sorts of different backgrounds were brought together in a way of life that was literally uniform. service rates for men born in the early 1920s approached 80%. and working toward a common goal, often under stress, brought them still closer together.

though strictly speaking world war ii lasted less than 4 years for the us, its effects lasted longer. wars make central governments more powerful, and world war ii was an extreme case of this. in the us, as in all the other allied countries, the federal government was slow to give up the new powers it had acquired. indeed, in some respects the war didn't end in 1945; the enemy just switched to the soviet union. in tax rates, federal power, defense spending, conscription, and nationalism, the decades after the war looked more like wartime than prewar peacetime. [3] and the social effects lasted too. the kid pulled into the army from behind a mule team in west virginia didn't simply go back to the farm afterward. something else was waiting for him, something that looked a lot like the army.

if total war was the big political story of the 20th century, the big economic story was the rise of a new kind of company. and this too tended to produce both social and economic cohesion. [4]

the 20th century was the century of the big, national corporation. general electric, general foods, general motors. developments in finance, communications, transportation, and manufacturing enabled a new type of company whose goal was above all scale. version 1 of this world was low-res: a duplo world of a few giant companies dominating each big market. [5]

the late 19th and early 20th centuries had been a time of consolidation, led especially by j. p. morgan. thousands of companies run by their founders were merged into a couple hundred giant ones run by

professional managers. economies of scale ruled the day. it seemed to people at the time that this was the final state of things. john d. rockefeller said in 1880

> the day of combination is here to stay. individualism has gone, never to return.

he turned out to be mistaken, but he seemed right for the next hundred years.

the consolidation that began in the late 19th century continued for most of the 20th. by the end of world war ii, as michael lind writes, "the major sectors of the economy were either organized as government-backed cartels or dominated by a few oligopolistic corporations."

for consumers this new world meant the same choices everywhere, but only a few of them. when i grew up there were only 2 or 3 of most things, and since they were all aiming at the middle of the market there wasn't much to differentiate them.

one of the most important instances of this phenomenon was in tv. here there were 3 choices: nbc, cbs, and abc. plus public tv for eggheads and communists. the programs that the 3 networks offered were indistinguishable. in fact, here there was a triple pressure toward the center. if one show did try something daring, local affiliates in conservative markets would make them stop. plus since tvs were expensive, whole families watched the same shows together, so they had to be suitable for everyone.

and not only did everyone get the same thing, they got it at the same time. it's difficult to imagine now, but every night tens of millions of families would sit down together in front of their tv set watching the same show, at the same time, as their next door neighbors. what happens now with the super bowl used to happen every night. we were literally in sync. [6]

in a way mid-century tv culture was good. the view it gave of the world was like you'd find in a children's book, and it probably had something of the effect that (parents hope) children's books have in making people behave better. but, like children's books, tv was also misleading. dangerously misleading, for adults. in his autobiography, robert macneil talks of seeing gruesome images that had just come in from vietnam and thinking, we can't show these to families while they're having dinner.

i know how pervasive the common culture was, because i tried to opt out of it, and it was practically impossible to find alternatives. when i was 13 i realized, more from internal evidence than any outside source, that the ideas we were being fed on tv were crap, and i stopped watching it. [7] but it wasn't just tv. it seemed like everything around me was crap. the politicians all saying the same things, the consumer brands making almost identical products with different labels stuck on to indicate how prestigious they were meant to be, the balloon-frame houses with fake "colonial" skins, the cars with several feet of gratuitous metal on each end that started to fall apart after a couple years, the "red delicious" apples that were red but only nominally apples. and in retrospect, it *was* crap. [8]

but when i went looking for alternatives to fill this void, i found practically nothing. there was no internet then. the only place to look was in the chain bookstore in our local shopping mall. [9] there i found a copy of *the atlantic*. i wish i could say it became a gateway into a wider world, but in fact i found it boring and incomprehensible. like a kid tasting whisky for the first time and pretending to like it, i preserved that magazine as carefully as if it had been a book. i'm sure i still have it somewhere. but though it was evidence that there was, somewhere, a world that wasn't red delicious, i didn't find it till college.

it wasn't just as consumers that the big companies made us similar. they did as employers too. within companies there were powerful forces pushing people toward a single model of how to look and act.

ies. and the models of how to look and act varied little between companies. meaning everyone within this world was expected to seem more or less the same. and not just those in the corporate world, but also everyone who aspired to it — which in the middle of the 20th century meant most people who weren't already in it. for most of the 20th century, working-class people tried hard to look middle class. you can see it in old photos. few adults aspired to look dangerous in 1950.

but the rise of national corporations didn't just compress us culturally. it compressed us economically too, and on both ends.

along with giant national corporations, we got giant national labor unions. and in the mid 20th century the corporations cut deals with the unions where they paid over market price for labor. partly because the unions were monopolies. [10] partly because, as components of oligopolies themselves, the corporations knew they could safely pass the cost on to their customers, because their competitors would have to as well. and partly because in mid-century most of the giant companies were still focused on finding new ways to milk economies of scale. just as startups rightly pay aws a premium over the cost of running their own servers so they can focus on growth, many of the big national corporations were willing to pay a premium for labor. [11]

as well as pushing incomes up from the bottom, by overpaying unions, the big companies of the 20th century also pushed incomes down at the top, by underpaying their top management. economist j. k. galbraith wrote in 1967 that "there are few corporations in which it would be suggested that executive salaries are at a maximum." [12]

to some extent this was an illusion. much of the de facto pay of executives never showed up on their income tax returns, because it took the form of perks. the higher the rate of income tax, the more pressure there was to pay employees upstream of it. (in the uk, where taxes were even higher than in the us, companies would even pay their kids' private school tuitions.) one of the most valuable things the big companies of the mid 20th century gave their employees was job security, and this too didn't show up in tax returns or income statistics. so the nature of employment in these organizations tended to yield falsely low numbers about economic inequality. but even accounting for that, the big companies paid their best people less than market price. there was no market; the expectation was that you'd work for the same company for decades if not your whole career. [13]

your work was so illiquid there was little chance of getting market price. but that same illiquidity also encouraged you not to seek it. if the company promised to employ you till you retired and give you a pension afterward, you didn't want to extract as much from it this year as you could. you needed to take care of the company so it could take care of you. especially when you'd been working with the same group of people for decades. if you tried to squeeze the company for more money, you were squeezing the organization that was going to take care of *them*. plus if you didn't put the company first you wouldn't be promoted, and if you couldn't switch ladders, promotion on this one was the only way up. [14]

to someone who'd spent several formative years in the armed forces, this situation didn't seem as strange as it does to us now. from their point of view, as big company executives, they were high-ranking officers. they got paid a lot more than privates. they got to have expense account lunches at the best restaurants and fly around on the company's gulfstreams. it probably didn't occur to most of them to ask if they were being paid market price.

the ultimate way to get market price is to work for yourself, by starting your own company. that seems obvious to any ambitious person now. but in the mid 20th century it was an alien concept. not because

starting one's own company seemed too ambitious, but because it didn't seem ambitious enough. even as late as the 1970s, when i grew up, the ambitious plan was to get lots of education at prestigious institutions, and then join some other prestigious institution and work one's way up the hierarchy. your prestige was the prestige of the institution you belonged to. people did start their own businesses of course, but educated people rarely did, because in those days there was practically zero concept of starting what we now call a startup: a business that starts small and grows big. that was much harder to do in the mid 20th century. starting one's own business meant starting a business that would start small and stay small. which in those days of big companies often meant scurrying around trying to avoid being trampled by elephants. it was more prestigious to be one of the executive class riding the elephant.

by the 1970s, no one stopped to wonder where the big prestigious companies had come from in the first place. it seemed like they'd always been there, like the chemical elements. and indeed, there was a double wall between ambitious kids in the 20th century and the origins of the big companies. many of the big companies were roll-ups that didn't have clear founders. and when they did, the founders didn't seem like us. nearly all of them had been uneducated, in the sense of not having been to college. they were what shakespeare called rude mechanicals. college trained one to be a member of the professional classes. its graduates didn't expect to do the sort of grubby menial work that andrew carnegie or henry ford started out doing. [15]

and in the 20th century there were more and more college graduates. they increased from about 2% of the population in 1900 to about 25% in 2000. in the middle of the century our two big forces intersect, in the form of the gi bill, which sent 2.2 million world war ii veterans to college. few thought of it in these terms, but the result of making college the canonical path for the ambitious was a world in which it was socially acceptable to work for henry ford, but not to be henry ford. [16]

i remember this world well. i came of age just as it was starting to break up. in my childhood it was still dominant. not quite so dominant as it had been. we could see from old tv shows and yearbooks and the way adults acted that people in the 1950s and 60s had been even more conformist than us. the mid-century model was already starting to get old. but that was not how we saw it at the time. we would at most have said that one could be a bit more daring in 1975 than 1965. and indeed, things hadn't changed much yet.

but change was coming soon. and when the duplo economy started to disintegrate, it disintegrated in several different ways at once. vertically integrated companies literally dis-integrated because it was more efficient to. incumbents faced new competitors as (a) markets went global and (b) technical innovation started to trump economies of scale, turning size from an asset into a liability. smaller companies were increasingly able to survive as formerly narrow channels to consumers broadened. markets themselves started to change faster, as whole new categories of products appeared. and last but not least, the federal government, which had previously smiled upon j. p. morgan's world as the natural state of things, began to realize it wasn't the last word after all.

what j. p. morgan was to the horizontal axis, henry ford was to the vertical. he wanted to do everything himself. the giant plant he built at river rouge between 1917 and 1928 literally took in iron ore at one end and sent cars out the other. 100,000 people worked there. at the time it seemed the future. but that is not how car companies operate today. now much of the design and manufacturing happens in a long supply chain, whose products the car companies ultimately assemble and sell. the reason car companies operate this way is that it works better. each company in the supply chain focuses on what they know best. and they each have to do it well or they can be swapped out for another supplier.

why didn't henry ford realize that networks of cooperating companies work better than a single big company? one reason is that supplier networks take a while to evolve. in 1917, doing everything himself seemed to ford the only way to get the scale he needed. and the second reason is that if you want to solve a problem using a network of cooperating companies, you have to be able to coordinate their efforts, and you can do that much better with computers. computers reduce the transaction costs that coase argued are the raison d'etre of corporations. that is a fundamental change.

in the early 20th century, big companies were synonymous with efficiency. in the late 20th century they were synonymous with inefficiency. to some extent this was because the companies themselves had become sclerotic. but it was also because our standards were higher.

it wasn't just within existing industries that change occurred. the industries themselves changed. it became possible to make lots of new things, and sometimes the existing companies weren't the ones who did it best.

microcomputers are a classic example. the market was pioneered by upstarts like apple. when it got big enough, ibm decided it was worth paying attention to. at the time ibm completely dominated the computer industry. they assumed that all they had to do, now that this market was ripe, was to reach out and pick it. most people at the time would have agreed with them. but what happened next illustrated how much more complicated the world had become. ibm did launch a microcomputer. though quite successful, it did not crush apple. but even more importantly, ibm itself ended up being supplanted by a supplier coming in from the side — from software, which didn't even seem to be the same business. ibm's big mistake was to accept a non-exclusive license for dos. it must have seemed a safe move at the time. no other computer manufacturer had ever been able to outsell them. what difference did it make if other manufacturers could offer dos too? the result of that miscalculation was an explosion of inexpensive pc clones. microsoft now owned the pc standard, and the customer. and the microcomputer business ended up being apple vs microsoft.

basically, apple bumped ibm and then microsoft stole its wallet. that sort of thing did not happen to big companies in mid-century. but it was going to happen increasingly often in the future.

change happened mostly by itself in the computer business. in other industries, legal obstacles had to be removed first. many of the mid-century oligopolies had been anointed by the federal government with policies (and in wartime, large orders) that kept out competitors. this didn't seem as dubious to government officials at the time as it sounds to us. they felt a two-party system ensured sufficient competition in politics. it ought to work for business too.

gradually the government realized that anti-competitive policies were doing more harm than good, and during the carter administration it started to remove them. the word used for this process was misleadingly narrow: deregulation. what was really happening was de-oligopolization. it happened to one industry after another. two of the most visible to consumers were air travel and long-distance phone service, which both became dramatically cheaper after deregulation.

deregulation also contributed to the wave of hostile takeovers in the 1980s. in the old days the only limit on the inefficiency of companies, short of actual bankruptcy, was the inefficiency of their competitors. now companies had to face absolute rather than relative standards. any public company that didn't generate sufficient returns on its assets risked having its management replaced with one that would. often the new managers did this by breaking companies up into components that were more valuable separately. [17]

version 1 of the national economy consisted of a few big blocks whose relationships were negotiated in back rooms by a handful of executives, politicians, regulators, and labor leaders. version 2 was higher resolution: there were more companies, of more different sizes, making more different things, and their relationships changed faster. in this world there were still plenty of back room negotiations, but more was left to market forces. which further accelerated the fragmentation.

it's a little misleading to talk of versions when describing a gradual process, but not as misleading as it might seem. there was a lot of change in a few decades, and what we ended up with was qualitatively different. the companies in the s&p 500 in 1958 had been there an average of 61 years. by 2012 that number was 18 years. [18]

the breakup of the duplo economy happened simultaneously with the spread of computing power. to what extent were computers a precondition? it would take a book to answer that. obviously the spread of computing power was a precondition for the rise of startups. i suspect it was for most of what happened in finance too. but was it a precondition for globalization or the lbo wave? i don't know, but i wouldn't discount the possibility. it may be that the refragmentation was driven by computers in the way the industrial revolution was driven by steam engines. whether or not computers were a precondition, they have certainly accelerated it.

the new fluidity of companies changed people's relationships with their employers. why climb a corporate ladder that might be yanked out from under you? ambitious people started to think of a career less as climbing a single ladder than as a series of jobs that might be at different companies. more movement (or even potential movement) between companies introduced more competition in salaries. plus as companies became smaller it became easier to estimate how much an employee contributed to the company's revenue. both changes drove salaries toward market price. and since people vary dramatically in productivity, paying market price meant salaries started to diverge.

by no coincidence it was in the early 1980s that the term "yuppie" was coined. that word is not much used now, because the phenomenon it describes is so taken for granted, but at the time it was a label for something novel. yuppies were young professionals who made lots of money. to someone in their twenties today, this wouldn't seem worth naming. why wouldn't young professionals make lots of money? but until the 1980s, being underpaid early in your career was part of what it meant to be a professional. young professionals were paying their dues, working their way up the ladder. the rewards would come later. what was novel about yuppies was that they wanted market price for the work they were doing now.

the first yuppies did not work for startups. that was still in the future. nor did they work for big companies. they were professionals working in fields like law, finance, and consulting. but their example rapidly inspired their peers. once they saw that new bmw 325i, they wanted one too.

underpaying people at the beginning of their career only works if everyone does it. once some employer breaks ranks, everyone else has to, or they can't get good people. and once started this process spreads through the whole economy, because at the beginnings of people's careers they can easily switch not merely employers but industries.

but not all young professionals benefitted. you had to produce to get paid a lot. it was no coincidence that the first yuppies worked in fields where it was easy to measure that.

more generally, an idea was returning whose name sounds old-fashioned precisely because it was so rare for so long: that you could make your fortune. as in the past there were multiple ways to do it. some made their fortunes by creating wealth, and others by playing zero-sum games. but once it bec-

ame possible to make one's fortune, the ambitious had to decide whether or not to. a physicist who chose physics over wall street in 1990 was making a sacrifice that a physicist in 1960 didn't have to think about.

the idea even flowed back into big companies. ceos of big companies make more now than they used to, and i think much of the reason is prestige. in 1960, corporate ceos had immense prestige. they were the winners of the only economic game in town. but if they made as little now as they did then, in real dollar terms, they'd seem like small fry compared to professional athletes and whiz kids making millions from startups and hedge funds. they don't like that idea, so now they try to get as much as they can, which is more than they had been getting. [19]

meanwhile a similar fragmentation was happening at the other end of the economic scale. as big companies' oligopolies became less secure, they were less able to pass costs on to customers and thus less willing to overpay for labor. and as the duplo world of a few big blocks fragmented into many companies of different sizes — some of them overseas — it became harder for unions to enforce their monopolies. as a result workers' wages also tended toward market price. which (inevitably, if unions had been doing their job) tended to be lower. perhaps dramatically so, if automation had decreased the need for some kind of work.

and just as the mid-century model induced social as well as economic cohesion, its breakup brought social as well as economic fragmentation. people started to dress and act differently. those who would later be called the "creative class" became more mobile. people who didn't care much for religion felt less pressure to go to church for appearances' sake, while those who liked it a lot opted for increasingly colorful forms. some switched from meat loaf to tofu, and others to hot pockets. some switched from driving ford sedans to driving small imported cars, and others to driving suvs. kids who went to private schools or wished they did started to dress "preppy," and kids who wanted to seem rebellious made a conscious effort to look disreputable. in a hundred ways people spread apart. [20]

almost four decades later, fragmentation is still increasing. has it been net good or bad? i don't know; the question may be unanswerable. not entirely bad though. we take for granted the forms of fragmentation we like, and worry only about the ones we don't. but as someone who caught the tail end of mid-century conformism, i can tell you it was no utopia. [21]

my goal here is not to say whether fragmentation has been good or bad, just to explain why it's happening. with the centripetal forces of total war and 20th century oligopoly mostly gone, what will happen next? and more specifically, is it possible to reverse some of the fragmentation we've seen?

if it is, it will have to happen piecemeal. you can't reproduce mid-century cohesion the way it was originally produced. it would be insane to go to war just to induce more national unity. and once you understand the degree to which the economic history of the 20th century was a low-res version 1, it's clear you can't reproduce that either.

20th century cohesion was something that happened at least in a sense naturally. the war was due mostly to external forces, and the duplo economy was an evolutionary phase. if you want cohesion now, you'd have to induce it deliberately. and it's not obvious how. i suspect the best we'll be able to do is address the symptoms of fragmentation. but that may be enough.

the form of fragmentation people worry most about lately is economic inequality, and if you want to eliminate that you're up against a truly formidable headwind that has been in operation since the stone age. technology.

technology is a lever. it magnifies work. and the lever not only grows increasingly long, but the rate at which it grows is itself increasing.

which in turn means the variation in the amount of wealth people can create has not only been increasing, but accelerating. the unusual conditions that prevailed in the mid 20th century masked this underlying trend. the ambitious had little choice but to join large organizations that made them march in step with lots of other people — literally in the case of the armed forces, figuratively in the case of big corporations. even if the big corporations had wanted to pay people proportionate to their value, they couldn't have figured out how. but that constraint has gone now. ever since it started to erode in the 1970s, we've seen the underlying forces at work again. [22]

not everyone who gets rich now does it by creating wealth, certainly. but a significant number do, and the baumol effect means all their peers get dragged along too. [23] and as long as it's possible to get rich by creating wealth, the default tendency will be for economic inequality to increase. even if you eliminate all the other ways to get rich. you can mitigate this with subsidies at the bottom and taxes at the top, but unless taxes are high enough to discourage people from creating wealth, you're always going to be fighting a losing battle against increasing variation in productivity. [24]

that form of fragmentation, like the others, is here to stay. or rather, back to stay. nothing is forever, but the tendency toward fragmentation should be more forever than most things, precisely because it's not due to any particular cause. it's simply a reversion to the mean. when rockefeller said individualism was gone, he was right for a hundred years. it's back now, and that's likely to be true for longer.

i worry that if we don't acknowledge this, we're headed for trouble. if we think 20th century cohesion disappeared because of few policy tweaks, we'll be deluded into thinking we can get it back (minus the bad parts, somehow) with a few countertweaks. and then we'll waste our time trying to eliminate fragmentation, when we'd be better off thinking about how to mitigate its consequences.

---

notes

1 lester thurow, writing in 1975, said the wage differentials prevailing at the end of world war ii had become so embedded that they "were regarded as 'just' even after the egalitarian pressures of world war ii had disappeared. basically, the same differentials exist to this day, thirty years later." but goldin and margo think market forces in the postwar period also helped preserve the wartime compression of wages — specifically increased demand for unskilled workers, and oversupply of educated ones.

(oddly enough, the american custom of having employers pay for health insurance derives from efforts by businesses to circumvent nwlb wage controls in order to attract workers.)

2 as always, tax rates don't tell the whole story. there were lots of exemptions, especially for individuals. and in world war ii the tax codes were so new that the government had little acquired immunity to tax avoidance. if the rich paid high taxes during the war it was more because they wanted to than because they had to.

after the war, federal tax receipts as a percentage of gdp were about the same as they are now. in fact, for the entire period since the war, tax receipts have stayed close to 18% of gdp, despite dramatic changes in tax rates. the lowest point occurred when marginal income tax rates were highest: 14.1% in 1950. looking at the data, it's hard to avoid the conclusion that tax rates have had little effect on what people actually paid.

3 though in fact the decade preceding the war had been a time of unprecedented federal power, in response to the depression. which is not entirely a coincidence, because the depression was one of the causes of the war. in many ways the new deal was a sort of dress rehearsal for the measures the federal government took during wartime. the wartime versions were much more drastic and more pervasive though. as anthony badger wrote, "for many americans the decisive change in their experiences came not with the new deal but with world war ii."

4 i don't know enough about the origins of the world wars to say, but it's not inconceivable they were connected to the rise of big corporations. if that were the case, 20th century cohesion would have a single cause.

5 more precisely, there was a bimodal economy consisting, in galbraith's words, of "the world of the technically dynamic, massively capitalized and highly organized corporations on the one hand and the hundreds of thousands of small and traditional proprietors on the other." money, prestige, and power were concentrated in the former, and there was near zero crossover.

6 i wonder how much of the decline in families eating together was due to the decline in families watching tv together afterward.

7 i know when this happened because it was the season dallas premiered. everyone else was talking about what was happening on dallas, and i had no idea what they meant.

8 i didn't realize it till i started doing research for this essay, but the meretriciousness of the products i grew up with is a well-known byproduct of oligopoly. when companies can't compete on price, they compete on tailfins.

9 monroeville mall was at the time of its completion in 1969 the largest in the country. in the late 1970s the movie *dawn of the dead* was shot there. apparently the mall was not just the location of the movie, but its inspiration; the crowds of shoppers drifting through this huge mall reminded george romero of zombies. my first job was scooping ice cream in the baskin-robbins.

10 labor unions were exempted from antitrust laws by the clayton antitrust act in 1914 on the grounds that a person's work is not "a commodity or article of commerce." i wonder if that means service companies are also exempt.

11 the relationships between unions and unionized companies can even be symbiotic, because unions will exert political pressure to protect their hosts. according to michael lind, when politicians tried to attack the a&p supermarket chain because it was putting local grocery stores out of business, "a&p successfully defended itself by allowing the unionization of its workforce in 1938, thereby gaining organized labor as a constituency." i've seen this phenomenon myself: hotel unions are responsible for more of the political pressure against airbnb than hotel companies.

12 galbraith was clearly puzzled that corporate executives would work so hard to make money for other people (the shareholders) instead of themselves. he devoted much of *the new industrial state* to trying to figure this out.

his theory was that professionalism had replaced money as a motive, and that modern corporate executives were, like (good) scientists, motivated less by financial rewards than by the desire to do good work and thereby earn the respect of their peers. there is something in this, though i think lack of movement between companies combined with self-interest explains much of observed behavior.

13 galbraith (p. 94) says a 1952 study of the 800 highest paid executives at 300 big corporations found that three quarters of them had been with their company for more than 20 years.

14 it seems likely that in the first third of the 20th century executive salaries were low partly because companies then were more dependent on banks, who would have disapproved if executives got too much. this was certainly true in the beginning. the first big company ceos were j. p. morgan's hired hands.

companies didn't start to finance themselves with retained earnings till the 1920s. till then they had to pay out their earnings in dividends, and so depended on banks for capital for expansion. bankers continued to sit on corporate boards till the glass-steagall act in 1933.

by mid-century big companies funded 3/4 of their growth from earnings. but the early years of bank dependence, reinforced by the financial controls of world war ii, must have had a big effect on social conventions about executive salaries. so it may be that the lack of movement between companies was as much the effect of low salaries as the cause.

incidentally, the switch in the 1920s to financing growth with retained earnings was one cause of the 1929 crash. the banks now had to find someone else to lend to, so they made more margin loans.

15 even now it's hard to get them to. one of the things i find hardest to get into the heads of would-be startup founders is how important it is to do certain kinds of menial work early in the life of a company. doing things that don't scale is to how henry ford got started as a high-fiber diet is to the traditional peasant's diet: they had no choice but to do the right thing, while we have to make a conscious effort.

16 founders weren't celebrated in the press when i was a kid. "our founder" meant a photograph of a severe-looking man with a walrus mustache and a wing collar who had died decades ago. the thing to be when i was a kid was an *executive*. if you weren't around then it's hard to grasp the cachet that term had. the fancy version of everything was called the "executive" model.

17 the wave of hostile takeovers in the 1980s was enabled by a combination of circumstances: court decisions striking down state anti-takeover laws, starting with the supreme court's 1982 decision in edgar v. mite corp.; the reagan administration's comparatively sympathetic attitude toward takeovers; the depository institutions act of 1982, which allowed banks and savings and loans to buy corporate bonds; a new sec rule issued in 1982 (rule 415) that made it possible to bring corporate bonds to market faster; the creation of the junk bond business by michael milken; a vogue for conglomerates in the preceding period that caused many companies to be combined that never should have been; a decade of inflation that left many public companies trading below the value of their assets; and not least, the increasing complacency of managements.

18 foster, richard. "creative destruction whips through corporate america." innosight, february 2012.

19 ceos of big companies may be overpaid. i don't know enough about big companies to say. but it is certainly not impossible for a ceo to make 200x as much difference to a company's revenues as the average employee. look at what steve jobs did for apple when he came back as ceo. it would have been a good deal for the board to give him 95% of the company. apple's market cap the day steve came back in july 1997 was 1.73 billion. 5% of apple now (january 2016) would be worth about 30 billion. and it would not be if steve hadn't come back; apple probably wouldn't even exist anymore.

merely including steve in the sample might be enough to answer the question of whether public company ceos in the aggregate are overpaid. and that is not as facile a trick as it might seem, because the broader your holdings, the more the aggregate is what you care about.

20 the late 1960s were famous for social upheaval. but that was more rebellion (which can happen in any era if people are provoked sufficiently) than fragmentation. you're not seeing fragmentation unless you see people breaking off to both left and right.

21 globally the trend has been in the other direction. while the us is becoming more fragmented, the world as a whole is becoming less fragmented, and mostly in good ways.

22 there were a handful of ways to make a fortune in the mid 20th century. the main one was drilling for oil, which was open to newcomers because it was not something big companies could dominate through economies of scale. how did individuals accumulate large fortunes in an era of such high taxes? giant tax loopholes defended by two of the most powerful men in congress, sam rayburn and lyndon johnson.

but becoming a texas oilman was not in 1950 something one could aspire to the way starting a startup or going to work on wall street were in 2000, because (a) there was a strong local component and (b) success depended so much on luck.

23 the baumol effect induced by startups is very visible in silicon valley. google will pay people millions of dollars a year to keep them from leaving to start or join startups.

24 i'm not claiming variation in productivity is the only cause of economic inequality in the us. but it's a significant cause, and it will become as big a cause as it needs to, in the sense that if you ban other ways to get rich, people who want to get rich will use this route instead.

thanks to sam altman, trevor blackwell, paul buchheit, patrick collison, ron conway, chris dixon, benedict evans, richard florida, ben horowitz, jessica livingston, robert morris, tim o'reilly, geoff ralston, max roser, alexia tsotsis, and qasar younis for reading drafts of this. max also told me about several valuable sources.

bibliography

allen, frederick lewis. *the big change*. harper, 1952.

averitt, robert. *the dual economy*. norton, 1968.

badger, anthony. *the new deal*. hill and wang, 1989.

bainbridge, john. *the super-americans*. doubleday, 1961.

beatty, jack. *collossus*. broadway, 2001.

brinkley, douglas. *wheels for the world*. viking, 2003.

brownleee, w. elliot. *federal taxation in america*. cambridge, 1996.

chandler, alfred. *the visible hand*. harvard, 1977.

chernow, ron. *the house of morgan*. simon & schuster, 1990.

chernow, ron. *titan: the life of john d. rockefeller*. random house, 1998.

galbraith, john. *the new industrial state*. houghton mifflin, 1967.

goldin, claudia and robert a. margo. "the great compression: the wage structure in the united states at mid-century." nber working paper 3817, 1991.

gordon, john. *an empire of wealth*. harpercollins, 2004.

klein, maury. *the genesis of industrial america, 1870-1920*. cambridge, 2007.

lind, michael. *land of promise*. harpercollins, 2012.

mickelthwaite, john, and adrian wooldridge. *the company*. modern library, 2003.

nasaw, david. *andrew carnegie*. penguin, 2006.

sobel, robert. *the age of giant corporations*. praeger, 1993.

thurow, lester. *generating inequality: mechanisms of distribution*. basic books, 1975.

witte, john. *the politics and development of the federal income tax*. wisconsin, 1985.

related:

07

cities and ambition

may 2008
paulgraham.com/cities.html

great cities attract ambitious people. you can sense it when you walk around one. in a hundred subtle ways, the city sends you a message: you could do more; you should try harder.

the surprising thing is how different these messages can be. new york tells you, above all: you should make more money. there are other messages too, of course. you should be hipper. you should be better looking. but the clearest message is that you should be richer.

what i like about boston (or rather cambridge) is that the message there is: you should be smarter. you really should get around to reading all those books you've been meaning to.

when you ask what message a city sends, you sometimes get surprising answers. as much as they respect brains in silicon valley, the message the valley sends is: you should be more powerful.

that's not quite the same message new york sends. power matters in new york too of course, but new york is pretty impressed by a billion dollars even if you merely inherited it. in silicon valley no one would care except a few real estate agents. what matters in silicon valley is how much effect you have on the world. the reason people there care about larry and sergey is not their wealth but the fact that they control google, which affects practically everyone.

———

how much does it matter what message a city sends? empirically, the answer seems to be: a lot. you might think that if you had enough strength of mind to do great things, you'd be able to transcend your environment. where you live should make at most a couple percent difference. but if you look at the historical evidence, it seems to matter more than that. most people who did great things were clumped together in a few places where that sort of thing was done at the time.

you can see how powerful cities are from something i wrote about earlier: the case of the milanese leonardo. practically every fifteenth century italian painter you've heard of was from florence, even though milan was just as big. people in florence weren't genetically different, so you have to assume there was someone born in milan with as much natural ability as leonardo. what happened to him?

if even someone with the same natural ability as leonardo couldn't beat the force of environment, do you suppose you can?

i don't. i'm fairly stubborn, but i wouldn't try to fight this force. i'd rather use it. so i've thought a lot about where to live.

i'd always imagined berkeley would be the ideal place — that it would basically be cambridge with good weather. but when i finally tried living there a couple years ago, it turned out not to be. the message berkeley sends is: you should live better. life in berkeley is very civilized. it's probably the place in america where someone from northern europe would feel most at home. but it's not humming with ambition.

in retrospect it shouldn't have been surprising that a place so pleasant would attract people interested above all in quality of life. cambridge with good weather, it turns out, is not cambridge. the people you find in cambridge are not there by accident. you have to make sacrifices to live there. it's expensive and somewhat grubby, and the weather's often bad. so the kind of people you find in cambridge are the kind of people who want to live where the smartest people are, even if that means living in an expensive, grubby place with bad weather.

as of this writing, cambridge seems to be the intellectual capital of the world. i realize that seems a preposterous claim. what makes it true is that it's more preposterous to claim about anywhere else.

american universities currently seem to be the best, judging from the flow of ambitious students. and what us city has a stronger claim? new york? a fair number of smart people, but diluted by a much larger number of neanderthals in suits. the bay area has a lot of smart people too, but again, diluted; there are two great universities, but they're far apart. harvard and mit are practically adjacent by west coast standards, and they're surrounded by about 20 other colleges and universities. [1]

cambridge as a result feels like a town whose main industry is ideas, while new york's is finance and silicon valley's is startups.

---

when you talk about cities in the sense we are, what you're really talking about is collections of people. for a long time cities were the only large collections of people, so you could use the two ideas interchangeably. but we can see how much things are changing from the examples i've mentioned. new york is a classic great city. but cambridge is just part of a city, and silicon valley is not even that. (san jose is not, as it sometimes claims, the capital of silicon valley. it's just 178 square miles at one end of it.)

maybe the internet will change things further. maybe one day the most important community you belong to will be a virtual one, and it won't matter where you live physically. but i wouldn't bet on it. the physical world is very high bandwidth, and some of the ways cities send you messages are quite subtle.

one of the exhilarating things about coming back to cambridge every spring is walking through the streets at dusk, when you can see into the houses. when you walk through palo alto in the evening, you see nothing but the blue glow of tvs. in cambridge you see shelves full of promising-looking books. palo alto was probably much like cambridge in 1960, but you'd never guess now that there was a university nearby. now it's just one of the richer neighborhoods in silicon valley. [2]

a city speaks to you mostly by accident — in things you see through windows, in conversations you overhear. it's not something you have to seek out, but something you can't turn off. one of the occupational hazards of living in cambridge is overhearing the conversations of people who use interrogative intonation in declarative sentences. but on average i'll take cambridge conversations over new york or silicon valley ones.

a friend who moved to silicon valley in the late 90s said the worst thing about living there was the low quality of the eavesdropping. at the time i thought she was being deliberately eccentric. sure, it can be interesting to eavesdrop on people, but is good quality eavesdropping so important that it would affect where you chose to live? now i understand what she meant. the conversations you overhear tell you what sort of people you're among.

---

no matter how determined you are, it's hard not to be influenced by the people around you. it's not so much that you do whatever a city expects of you, but that you get discouraged when no one around you cares about the same things you do.

there's an imbalance between encouragement and discouragement like that between gaining and losing money. most people overvalue negative amounts of money: they'll work much harder to avoid losing a dollar than to gain one. similarly, although there are plenty of people strong enough to resist doing something just because that's what one is supposed to do where they happen to be, there are few strong enough to keep working on something no one around them cares about.

because ambitions are to some extent incompatible and admiration is a zero-sum game, each city tends to focus on one type of ambition. the reason cambridge is the intellectual capital is not just that there's

a concentration of smart people there, but that there's nothing *else* people there care about more. professors in new york and the bay area are second class citizens — till they start hedge funds or startups respectively.

this suggests an answer to a question people in new york have wondered about since the bubble: whether new york could grow into a startup hub to rival silicon valley. one reason that's unlikely is that someone starting a startup in new york would feel like a second class citizen. [3] there's already something else people in new york admire more.

in the long term, that could be a bad thing for new york. the power of an important new technology does eventually convert to money. so by caring more about money and less about power than silicon valley, new york is recognizing the same thing, but slower. [4] and in fact it has been losing to silicon valley at its own game: the ratio of new york to california residents in the forbes 400 has decreased from 1.45 (81:56) when the list was first published in 1982 to .83 (73:88) in 2007.

———

not all cities send a message. only those that are centers for some type of ambition do. and it can be hard to tell exactly what message a city sends without living there. i understand the messages of new york, cambridge, and silicon valley because i've lived for several years in each of them. dc and la seem to send messages too, but i haven't spent long enough in either to say for sure what they are.

the big thing in la seems to be fame. there's an a list of people who are most in demand right now, and what's most admired is to be on it, or friends with those who are. beneath that, the message is much like new york's, though perhaps with more emphasis on physical attractiveness.

in dc the message seems to be that the most important thing is who you know. you want to be an insider. in practice this seems to work much as in la. there's an a list and you want to be on it or close to those who are. the only difference is how the a list is selected. and even that is not that different.

at the moment, san francisco's message seems to be the same as berkeley's: you should live better. but this will change if enough startups choose sf over the valley. during the bubble that was a predictor of failure — a self-indulgent choice, like buying expensive office furniture. even now i'm suspicious when startups choose sf. but if enough good ones do, it stops being a self-indulgent choice, because the center of gravity of silicon valley will shift there.

i haven't found anything like cambridge for intellectual ambition. oxford and cambridge (england) feel like ithaca or hanover: the message is there, but not as strong.

paris was once a great intellectual center. if you went there in 1300, it might have sent the message cambridge does now. but i tried living there for a bit last year, and the ambitions of the inhabitants are not intellectual ones. the message paris sends now is: do things with style. i liked that, actually. paris is the only city i've lived in where people genuinely cared about art. in america only a few rich people buy original art, and even the more sophisticated ones rarely get past judging it by the brand name of the artist. but looking through windows at dusk in paris you can see that people there actually care what paintings look like. visually, paris has the best eavesdropping i know. [5]

there's one more message i've heard from cities: in london you can still (barely) hear the message that one should be more aristocratic. if you listen for it you can also hear it in paris, new york, and boston. but this message is everywhere very faint. it would have been strong 100 years ago, but now i probably wouldn't have picked it up at all if i hadn't deliberately tuned in to that wavelength to see if there was any signal left.

---

so far the complete list of messages i've picked up from cities is: wealth, style, hipness, physical attractiveness, fame, political power, economic power, intelligence, social class, and quality of life.

my immediate reaction to this list is that it makes me slightly queasy. i'd always considered ambition a good thing, but i realize now that was because i'd always implicitly understood it to mean ambition in the areas i cared about. when you list everything ambitious people are ambitious about, it's not so pretty.

on closer examination i see a couple things on the list that are surprising in the light of history. for example, physical attractiveness wouldn't have been there 100 years ago (though it might have been 2400 years ago). it has always mattered for women, but in the late twentieth century it seems to have started to matter for men as well. i'm not sure why — probably some combination of the increasing power of women, the increasing influence of actors as models, and the fact that so many people work in offices now: you can't show off by wearing clothes too fancy to wear in a factory, so you have to show off with your body instead.

hipness is another thing you wouldn't have seen on the list 100 years ago. or wouldn't you? what it means is to know what's what. so maybe it has simply replaced the component of social class that consisted of being "au fait." that could explain why hipness seems particularly admired in london: it's version 2 of the traditional english delight in obscure codes that only insiders understand.

economic power would have been on the list 100 years ago, but what we mean by it is changing. it used to mean the control of vast human and material resources. but increasingly it means the ability to direct the course of technology, and some of the people in a position to do that are not even rich — leaders of important open source projects, for example. the captains of industry of times past had laboratories full of clever people cooking up new technologies for them. the new breed are themselves those people.

as this force gets more attention, another is dropping off the list: social class. i think the two changes are related. economic power, wealth, and social class are just names for the same thing at different stages in its life: economic power converts to wealth, and wealth to social class. so the focus of admiration is simply shifting upstream.

---

does anyone who wants to do great work have to live in a great city? no; all great cities inspire some sort of ambition, but they aren't the only places that do. for some kinds of work, all you need is a handful of talented colleagues.

what cities provide is an audience, and a funnel for peers. these aren't so critical in something like math or physics, where no audience matters except your peers, and judging ability is sufficiently straightforward that hiring and admissions committees can do it reliably. in a field like math or physics all you need is a department with the right colleagues in it. it could be anywhere — in los alamos, new mexico, for example.

it's in fields like the arts or writing or technology that the larger environment matters. in these the best practitioners aren't conveniently collected in a few top university departments and research labs — partly because talent is harder to judge, and partly because people pay for these things, so one doesn't need to rely on teaching or research funding to support oneself. it's in these more chaotic fields that it helps most to be in a great city: you need the encouragement of feeling that people around you care

about the kind of work you do, and since you have to find peers for yourself, you need the much larger intake mechanism of a great city.

you don't have to live in a great city your whole life to benefit from it. the critical years seem to be the early and middle ones of your career. clearly you don't have to grow up in a great city. nor does it seem to matter if you go to college in one. to most college students a world of a few thousand people seems big enough. plus in college you don't yet have to face the hardest kind of work — discovering new problems to solve.

it's when you move on to the next and much harder step that it helps most to be in a place where you can find peers and encouragement. you seem to be able to leave, if you want, once you've found both. the impressionists show the typical pattern: they were born all over france (pissarro was born in the caribbean) and died all over france, but what defined them were the years they spent together in paris.

---

unless you're sure what you want to do and where the leading center for it is, your best bet is probably to try living in several places when you're young. you can never tell what message a city sends till you live there, or even whether it still sends one. often your information will be wrong: i tried living in florence when i was 25, thinking it would be an art center, but it turned out i was 450 years too late.

even when a city is still a live center of ambition, you won't know for sure whether its message will resonate with you till you hear it. when i moved to new york, i was very excited at first. it's an exciting place. so it took me quite a while to realize i just wasn't like the people there. i kept searching for the cambridge of new york. it turned out it was way, way uptown: an hour uptown by air.

some people know at 16 what sort of work they're going to do, but in most ambitious kids, ambition seems to precede anything specific to be ambitious about. they know they want to do something great. they just haven't decided yet whether they're going to be a rock star or a brain surgeon. there's nothing wrong with that. but it means if you have this most common type of ambition, you'll probably have to figure out where to live by trial and error. you'll probably have to find the city where you feel at home to know what sort of ambition you have.

---

notes

1 this is one of the advantages of not having the universities in your country controlled by the government. when governments decide how to allocate resources, political deal-making causes things to be spread out geographically. no central goverment would put its two best universities in the same town, unless it was the capital (which would cause other problems). but scholars seem to like to cluster together as much as people in any other field, and when given the freedom to they derive the same advantages from it.

2 there are still a few old professors in palo alto, but one by one they die and their houses are transformed by developers into mcmansions and sold to vps of bus dev.

3 how many times have you read about startup founders who continued to live inexpensively as their companies took off? who continued to dress in jeans and t-shirts, to drive the old car they had in grad school, and so on? if you did that in new york, people would treat you like shit. if you walk into a fancy restaurant in san francisco wearing a jeans and a t-shirt, they're nice to you; who knows who you might be? not in new york.

one sign of a city's potential as a technology center is the number of restaurants that still require jackets for men. according to zagat's there are none in san francisco, la, boston, or seattle, 4 in dc, 6 in

chicago, 8 in london, 13 in new york, and 20 in paris.

(zagat's lists the ritz carlton dining room in sf as requiring jackets but i couldn't believe it, so i called to check and in fact they don't. apparently there's only one restaurant left on the entire west coast that still requires jackets: the french laundry in napa valley.)

4 ideas are one step upstream from economic power, so it's conceivable that intellectual centers like cambridge will one day have an edge over silicon valley like the one the valley has over new york.

this seems unlikely at the moment; if anything boston is falling further and further behind. the only reason i even mention the possibility is that the path from ideas to startups has recently been getting smoother. it's a lot easier now for a couple of hackers with no business experience to start a startup than it was 10 years ago. if you extrapolate another 20 years, maybe the balance of power will start to shift back. i wouldn't bet on it, but i wouldn't bet against it either.

5 if paris is where people care most about art, why is new york the center of gravity of the art business? because in the twentieth century, art as brand split apart from art as stuff. new york is where the richest buyers are, but all they demand from art is brand, and since you can base brand on anything with a sufficiently identifiable style, you may as well use the local stuff.

thanks to trevor blackwell, sarah harlin, jessica livingston, jackie mcdonough, robert morris, and david sloo for reading drafts of this.

08

the four quadrants of conformism

july 2020
paulgraham.com/conformism.html

one of the most revealing ways to classify people is by the degree and aggressiveness of their conformism. imagine a cartesian coordinate system whose horizontal axis runs from conventional-minded on the left to independent-minded on the right, and whose vertical axis runs from passive at the bottom to aggressive at the top. the resulting four quadrants define four types of people. starting in the upper left and going counter-clockwise: aggressively conventional-minded, passively conventional-minded, passively independent-minded, and aggressively independent-minded.

i think that you'll find all four types in most societies, and that which quadrant people fall into depends more on their own personality than the beliefs prevalent in their society. [1]

young children offer some of the best evidence for both points. anyone who's been to primary school has seen the four types, and the fact that school rules are so arbitrary is strong evidence that which quadrant people fall into depends more on them than the rules.

the kids in the upper left quadrant, the aggressively conventional-minded ones, are the tattletales. they believe not only that rules must be obeyed, but that those who disobey them must be punished.

the kids in the lower left quadrant, the passively conventional-minded, are the sheep. they're careful to obey the rules, but when other kids break them, their impulse is to worry that those kids will be punished, not to ensure that they will.

the kids in the lower right quadrant, the passively independent-minded, are the dreamy ones. they don't care much about rules and probably aren't 100% sure what the rules even are.

and the kids in the upper right quadrant, the aggressively independent-minded, are the naughty ones. when they see a rule, their first impulse is to question it. merely being told what to do makes them inclined to do the opposite.

when measuring conformism, of course, you have to say with respect to what, and this changes as kids get older. for younger kids it's the rules set by adults. but as kids get older, the source of rules becomes their peers. so a pack of teenagers who all flout school rules in the same way are not independent-minded; rather the opposite.

in adulthood we can recognize the four types by their distinctive calls, much as you could recognize four species of birds. the call of the aggressively conventional-minded is "crush <outgroup>!" (it's rather alarming to see an exclamation point after a variable, but that's the whole problem with the aggressively conventional-minded.) the call of the passively conventional-minded is "what will the neighbors think?" the call of the passively independent-minded is "to each his own." and the call of the aggressively independent-minded is "eppur si muove."

the four types are not equally common. there are more passive people than aggressive ones, and far more conventional-minded people than independent-minded ones. so the passively conventional-minded are the largest group, and the aggressively independent-minded the smallest.

since one's quadrant depends more on one's personality than the nature of the rules, most people would occupy the same quadrant even if they'd grown up in a quite different society.

princeton professor robert george recently wrote:

> i sometimes ask students what their position on slavery would have been had they been white and living in the south before abolition. guess what? they all would have been abolitionists! they all would have bravely spoken out against slavery, and worked tirelessly against it.

he's too polite to say so, but of course they wouldn't. and indeed, our default assumption should not merely be that his students would, on average, have behaved the same way people did at the time, but that the ones who are aggressively conventional-minded today would have been aggressively conventional-minded then too. in other words, that they'd not only not have fought against slavery, but that they'd have been among its staunchest defenders.

i'm biased, i admit, but it seems to me that aggressively conventional-minded people are responsible for a disproportionate amount of the trouble in the world, and that a lot of the customs we've evolved since the enlightenment have been designed to protect the rest of us from them. in particular, the retirement of the concept of heresy and its replacement by the principle of freely debating all sorts of different ideas, even ones that are currently considered unacceptable, without any punishment for those who try them out to see if they work. [2]

why do the independent-minded need to be protected, though? because they have all the new ideas. to be a successful scientist, for example, it's not enough just to be right. you have to be right when everyone else is wrong. conventional-minded people can't do that. for similar reasons, all successful startup ceos are not merely independent-minded, but aggressively so. so it's no coincidence that societies prosper only to the extent that they have customs for keeping the conventional-minded at bay. [3]

in the last few years, many of us have noticed that the customs protecting free inquiry have been weakened. some say we're overreacting — that they haven't been weakened very much, or that they've been weakened in the service of a greater good. the latter i'll dispose of immediately. when the conventional-minded get the upper hand, they always say it's in the service of a greater good. it just happens to be a different, incompatible greater good each time.

as for the former worry, that the independent-minded are being oversensitive, and that free inquiry hasn't been shut down that much, you can't judge that unless you are yourself independent-minded. you can't know how much of the space of ideas is being lopped off unless you have them, and only the independent-minded have the ones at the edges. precisely because of this, they tend to be very sensitive to changes in how freely one can explore ideas. they're the canaries in this coalmine.

the conventional-minded say, as they always do, that they don't want to shut down the discussion of all ideas, just the bad ones.

you'd think it would be obvious just from that sentence what a dangerous game they're playing. but i'll spell it out. there are two reasons why we need to be able to discuss even "bad" ideas.

the first is that any process for deciding which ideas to ban is bound to make mistakes. all the more so because no one intelligent wants to undertake that kind of work, so it ends up being done by the stupid. and when a process makes a lot of mistakes, you need to leave a margin for error. which in this case means you need to ban fewer ideas than you'd like to. but that's hard for the aggressively conventional-minded to do, partly because they enjoy seeing people punished, as they have since they were children, and partly because they compete with one another. enforcers of orthodoxy can't allow a borderline idea to exist, because that gives other enforcers an opportunity to one-up them in the moral purity department, and perhaps even to turn enforcer upon them. so instead of getting the margin for error we need, we get the opposite: a race to the bottom in which any idea that seems at all bannable ends up being banned. [4]

the second reason it's dangerous to ban the discussion of ideas is that ideas are more closely related than they look. which means if you restrict the discussion of some topics, it doesn't only affect those topics. the restrictions propagate back into any topic that yields implications in the forbidden ones. and

that is not an edge case. the best ideas do exactly that: they have consequences in fields far removed from their origins. having ideas in a world where some ideas are banned is like playing soccer on a pitch that has a minefield in one corner. you don't just play the same game you would have, but on a different shaped pitch. you play a much more subdued game even on the ground that's safe.

in the past, the way the independent-minded protected themselves was to congregate in a handful of places — first in courts, and later in universities — where they could to some extent make their own rules. places where people work with ideas tend to have customs protecting free inquiry, for the same reason wafer fabs have powerful air filters, or recording studios good sound insulation. for the last couple centuries at least, when the aggressively conventional-minded were on the rampage for whatever reason, universities were the safest places to be.

that may not work this time though, due to the unfortunate fact that the latest wave of intolerance began in universities. it began in the mid 1980s, and by 2000 seemed to have died down, but it has recently flared up again with the arrival of social media. this seems, unfortunately, to have been an own goal by silicon valley. though the people who run silicon valley are almost all independent-minded, they've handed the aggressively conventional-minded a tool such as they could only have dreamed of.

on the other hand, perhaps the decline in the spirit of free inquiry within universities is as much the symptom of the departure of the independent-minded as the cause. people who would have become professors 50 years ago have other options now. now they can become quants or start startups. you have to be independent-minded to succeed at either of those. if these people had been professors, they'd have put up a stiffer resistance on behalf of academic freedom. so perhaps the picture of the independent-minded fleeing declining universities is too gloomy. perhaps the universities are declining because so many have already left. [5]

though i've spent a lot of time thinking about this situation, i can't predict how it plays out. could some universities reverse the current trend and remain places where the independent-minded want to congregate? or will the independent-minded gradually abandon them? i worry a lot about what we might lose if that happened.

but i'm hopeful long term. the independent-minded are good at protecting themselves. if existing institutions are compromised, they'll create new ones. that may require some imagination. but imagination is, after all, their specialty.

---

notes

1 i realize of course that if people's personalities vary in any two ways, you can use them as axes and call the resulting four quadrants personality types. so what i'm really claiming is that the axes are orthogonal and that there's significant variation in both.

2 the aggressively conventional-minded aren't responsible for all the trouble in the world. another big source of trouble is the sort of charismatic leader who gains power by appealing to them. they become much more dangerous when such leaders emerge.

3 i never worried about writing things that offended the conventional-minded when i was running y combinator. if yc were a cookie company, i'd have faced a difficult moral choice. conventional-minded people eat cookies too. but they don't start successful startups. so if i deterred them from applying to yc, the only effect was to save us work reading applications.

4 there has been progress in one area: the punishments for talking about banned ideas are less severe than in the past. there's little danger of being killed, at least in richer countries. the aggressively conventional-minded are mostly satisfied with getting people fired.

5 many professors are independent-minded — especially in math, the hard sciences, and engineering, where you have to be to succeed. but students are more representative of the general population, and thus mostly conventional-minded. so when professors and students are in conflict, it's not just a conflict between generations but also between different types of people.

thanks to sam altman, trevor blackwell, nicholas christakis, patrick collison, sam gichuru, jessica livingston, patrick mckenzie, geoff ralston, and harj taggar for reading drafts of this.

09

orthodox privilege

july 2020
paulgraham.com/orth.html

"few people are capable of expressing with equanimity opinions which differ from the prejudices of their social environment. most people are even incapable of forming such opinions."

— einstein

there has been a lot of talk about privilege lately. although the concept is overused, there is something to it, and in particular to the idea that privilege makes you blind — that you can't see things that are visible to someone whose life is very different from yours.

but one of the most pervasive examples of this kind of blindness is one that i haven't seen mentioned explicitly. i'm going to call it *orthodox privilege*: the more conventional-minded someone is, the more it seems to them that it's safe for everyone to express their opinions.

it's safe for *them* to express their opinions, because the source of their opinions is whatever it's currently acceptable to believe. so it seems to them that it must be safe for everyone. they literally can't imagine a true statement that would get you in trouble.

and yet at every point in history, there were true things that would get you in trouble to say. is ours the first where this isn't so? what an amazing coincidence that would be.

surely it should at least be the default assumption that our time is not unique, and that there are true things you can't say now, just as there have always been. you would think. but even in the face of such overwhelming historical evidence, most people will go with their gut on this one.

in the most extreme cases, people suffering from orthodox privilege will not only deny that there's anything true that you can't say, but will accuse you of heresy merely for saying there is. though if there's more than one heresy current in your time, these accusations will be weirdly non-deterministic: you must either be an xist or a yist.

frustrating as it is to deal with these people, it's important to realize that they're in earnest. they're not pretending they think it's impossible for an idea to be both unorthodox and true. the world really looks that way to them.

indeed, this is a uniquely tenacious form of privilege. people can overcome the blindness induced by most forms of privilege by learning more about whatever they're not. but they can't overcome orthodox privilege just by learning more. they'd have to become more independent-minded. if that happens at all, it doesn't happen on the time scale of one conversation.

it may be possible to convince some people that orthodox privilege must exist even though they can't sense it, just as one can with, say, dark matter. there may be some who could be convinced, for example, that it's very unlikely that this is the first point in history at which there's nothing true you can't say, even if they can't imagine specific examples.

but in general i don't think it will work to say "check your privilege" about this type of privilege, because those in its demographic don't realize they're in it. it doesn't seem to conventional-minded people that they're conventional-minded. it just seems to them that they're right. indeed, they tend to be particularly sure of it.

perhaps the solution is to appeal to politeness. if someone says they can hear a high-pitched noise that you can't, it's only polite to take them at their word, instead of demanding evidence that's impossible to produce, or simply denying that they hear anything. imagine how rude that would seem. similarly, if someone says they can think of things that are true but that cannot be said, it's only polite to take them at their word, even if you can't think of any yourself.

thanks to sam altman, trevor blackwell, patrick collison, antonio garcia-martinez, jessica livingston, robert morris, michael nielsen, geoff ralston, max roser, and harj taggar for reading drafts of this.

10

the two kinds of moderate

december 2019
paulgraham.com/mod.html

there are two distinct ways to be politically moderate: on purpose and by accident. intentional moderates are trimmers, deliberately choosing a position mid-way between the extremes of right and left. accidental moderates end up in the middle, on average, because they make up their own minds about each question, and the far right and far left are roughly equally wrong.

you can distinguish intentional from accidental moderates by the distribution of their opinions. if the far left opinion on some matter is 0 and the far right opinion 100, an intentional moderate's opinion on every question will be near 50. whereas an accidental moderate's opinions will be scattered over a broad range, but will, like those of the intentional moderate, average to about 50.

intentional moderates are similar to those on the far left and the far right in that their opinions are, in a sense, not their own. the defining quality of an ideologue, whether on the left or the right, is to acquire one's opinions in bulk. you don't get to pick and choose. your opinions about taxation can be predicted from your opinions about sex. and although intentional moderates might seem to be the opposite of ideologues, their beliefs (though in their case the word "positions" might be more accurate) are also acquired in bulk. if the median opinion shifts to the right or left, the intentional moderate must shift with it. otherwise they stop being moderate.

accidental moderates, on the other hand, not only choose their own answers, but choose their own questions. they may not care at all about questions that the left and right both think are terribly important. so you can only even measure the politics of an accidental moderate from the intersection of the questions they care about and those the left and right care about, and this can sometimes be vanishingly small.

it is not merely a manipulative rhetorical trick to say "if you're not with us, you're against us," but often simply false.

moderates are sometimes derided as cowards, particularly by the extreme left. but while it may be accurate to call intentional moderates cowards, openly being an accidental moderate requires the most courage of all, because you get attacked from both right and left, and you don't have the comfort of being an orthodox member of a large group to sustain you.

nearly all the most impressive people i know are accidental moderates. if i knew a lot of professional athletes, or people in the entertainment business, that might be different. being on the far left or far right doesn't affect how fast you run or how well you sing. but someone who works with ideas has to be independent-minded to do it well.

or more precisely, you have to be independent-minded about the ideas you work with. you could be mindlessly doctrinaire in your politics and still be a good mathematician. in the 20th century, a lot of very smart people were marxists — just no one who was smart about the subjects marxism involves. but if the ideas you use in your work intersect with the politics of your time, you have two choices: be an accidental moderate, or be mediocre.

---

notes

1 it's possible in theory for one side to be entirely right and the other to be entirely wrong. indeed, ideologues must always believe this is the case. but historically it rarely has been.

2 for some reason the far right tend to ignore moderates rather than despise them as backsliders. i'm not sure why. perhaps it means that the far right is less ideological than the far left. or perhaps that they are more conf-

ident, or more resigned, or simply more disorganized. i just don't know.

3 having heretical opinions doesn't mean you have to express them openly. it may be easier to have them if you don't.

thanks to austen allred, trevor blackwell, patrick collison, jessica livingston, amjad masad, ryan petersen, and harj taggar for reading drafts of this.

11

haters

january 2020
paulgraham.com/fh.html

(i originally intended this for startup founders, who are often surprised by the attention they get as their companies grow, but it applies equally to anyone who becomes famous.)

if you become sufficiently famous, you'll acquire some fans who like you too much. these people are sometimes called "fanboys," and though i dislike that term, i'm going to have to use it here. we need some word for them, because this is a distinct phenomenon from someone simply liking your work.

a fanboy is obsessive and uncritical. liking you becomes part of their identity, and they create an image of you in their own head that is much better than reality. everything you do is good, because you do it. if you do something bad, they find a way to see it as good. and their love for you is not, usually, a quiet, private one. they want everyone to know how great you are.

well, you may be thinking, i could do without this kind of obsessive fan, but i know there are all kinds of people in the world, and if this is the worst consequence of fame, that's not so bad.

unfortunately this is not the worst consequence of fame. as well as fanboys, you'll have haters.

a hater is obsessive and uncritical. disliking you becomes part of their identity, and they create an image of you in their own head that is much worse than reality. everything you do is bad, because you do it. if you do something good, they find a way to see it as bad. and their dislike for you is not, usually, a quiet, private one. they want everyone to know how awful you are.

if you're thinking of checking, i'll save you the trouble. the second and fifth paragraphs are identical except for "good" being switched to "bad" and so on.

i spent years puzzling about haters. what are they, and where do they come from? then one day it dawned on me. haters are just fanboys with the sign switched.

note that by haters, i don't simply mean trolls. i'm not talking about people who say bad things about you and then move on. i'm talking about the much smaller group of people for whom this becomes a kind of obsession and who do it repeatedly over a long period.

like fans, haters seem to be an automatic consequence of fame. anyone sufficiently famous will have them. and like fans, haters are energized by the fame of whoever they hate. they hear a song by some pop singer. they don't like it much. if the singer were an obscure one, they'd just forget about it. but instead they keep hearing her name, and this seems to drive some people crazy. everyone's always going on about this singer, but she's no good! she's a fraud!

that word "fraud" is an important one. it's the spectral signature of a hater to regard the object of their hatred as a fraud. they can't deny their fame. indeed, their fame is if anything exaggerated in the hater's mind. they notice every mention of the singer's name, because every mention makes them angrier. in their own minds they exaggerate both the singer's fame and her lack of talent, and the only way to reconcile those two ideas is to conclude that she has tricked everyone.

what sort of people become haters? can anyone become one? i'm not sure about this, but i've noticed some patterns. haters are generally losers in a very specific sense: although they are occasionally talented, they have never achieved much. and indeed, anyone successful enough to have achieved significant fame would be unlikely to regard another famous person as a fraud on that account, because anyone famous knows how random fame is.

but haters are not always complete losers. they are not always the proverbial guy living in his mom's basement. many are, but some have some amount of talent. in fact i suspect that a sense of frustrated

talent is what drives some people to become haters. they're not just saying "it's unfair that so-and-so is famous," but "it's unfair that so-and-so is famous, and not me."

could a hater be cured if they achieved something impressive? my guess is that's a moot point, because they never will. i've been able to observe for long enough that i'm fairly confident the pattern works both ways: not only do people who do great work never become haters, haters never do great work. although i dislike the word "fanboy," it's evocative of something important about both haters and fanboys. it implies that the fanboy is so slavishly predictable in his admiration that he's diminished as a result, that he's less than a man.

haters seem even more diminished. i can imagine being a fanboy. i can think of people whose work i admire so much that i could abase myself before them out of sheer gratitude. if p. g. wodehouse were still alive, i could see myself being a wodehouse fanboy. but i could not imagine being a hater.

knowing that haters are just fanboys with the sign bit flipped makes it much easier to deal with them. we don't need a separate theory of haters. we can just use existing techniques for dealing with obsessive fans.

the most important of which is simply not to think much about them. if you're like most people who become famous enough to acquire haters, your initial reaction will be one of mystification. why does this guy seem to have it in for me? where does his obsessive energy come from, and what makes him so appallingly nasty? what did i do to set him off? is it something i can fix?

the mistake here is to think of the hater as someone you have a dispute with. when you have a dispute with someone, it's usually a good idea to try to understand why they're upset and then fix things if you can. disputes are distracting. but it's a false analogy to think of a hater as someone you have a dispute with. it's an understandable mistake, if you've never encountered haters before. but when you realize that you're dealing with a hater, and what a hater is, it's clear that it's a waste of time even to think about them. if you have obsessive fans, do you spend any time wondering what makes them love you so much? no, you just think "some people are kind of crazy," and that's the end of it.

since haters are equivalent to fanboys, that's the way to deal with them too. there may have been something that set them off. but it's not something that would have set off a normal person, so there's no reason to spend any time thinking about it. it's not you, it's them.

---

notes

1 there are of course some people who are genuine frauds. how can you distinguish between x calling y a fraud because x is a hater, and because y is a fraud? look at neutral opinion. actual frauds are usually pretty conspicuous. thoughtful people are rarely taken in by them. so if there are some thoughtful people who like y, you can usually assume y is not a fraud.

2 i would make an exception for teenagers, who sometimes act in such extreme ways that they are literally not themselves. i can imagine a teenage kid being a hater and then growing out of it. but not anyone over 25.

3 i have a much worse memory for misdeeds than my wife jessica, who is a connoisseur of character, but i don't wish it were better. most disputes are a waste of time even if you're in the right, and it's easy to bury the hatchet with someone if you can't remember why you were mad at them.

4 a competent hater will not merely attack you individually but will try to get mobs after you. in some cases you may want to refute whatever bogus claim they made in order to do so. but err on the side of not, because

ately it probably won't matter.

thanks to austen allred, trevor blackwell, patrick collison, christine ford, daniel gackle, jessica livingston, robert morris, elon musk, harj taggar, and peter thiel for reading drafts of this.

12

the origins of wokeness

january 2025
paulgraham.com/woke.html

the word "prig" isn't very common now, but if you look up the definition, it will sound familiar. google's isn't bad:

> a self-righteously moralistic person who behaves as if superior to others.

this sense of the word originated in the 18th century, and its age is an important clue: it shows that although wokeness is a comparatively recent phenomenon, it's an instance of a much older one.

there's a certain kind of person who's attracted to a shallow, exacting kind of moral purity, and who demonstrates his purity by attacking anyone who breaks the rules. every society has these people. all that changes is the rules they enforce. in victorian england it was christian virtue. in stalin's russia it was orthodox marxism-leninism. for the woke, it's social justice.

so if you want to understand wokeness, the question to ask is not why people behave this way. every society has prigs. the question to ask is why our prigs are priggish about these ideas, at this moment. and to answer that we have to ask when and where wokeness began.

the answer to the first question is the 1980s. wokeness is a second, more aggressive wave of political correctness, which started in the late 1980s, died down in the late 1990s, and then returned with a vengeance in the early 2010s, finally peaking after the riots of 2020.

what was political correctness, exactly? i'm often asked to define both this term and wokeness by people who think they're meaningless labels, so i will. they both have the same definition:

> an aggressively performative focus on social justice.

in other words, it's people being prigs about social justice. and that's the real problem — the performativeness, not the social justice. [0]

racism, for example, is a genuine problem. not a problem on the scale that the woke believe it to be, but a genuine one. i don't think any reasonable person would deny that. the problem with political correctness was not that it focused on marginalized groups, but the shallow, aggressive way in which it did so. instead of going out into the world and quietly helping members of marginalized groups, the politically correct focused on getting people in trouble for using the wrong words to talk about them.

as for where political correctness began, if you think about it, you probably already know the answer. did it begin outside universities and spread to them from this external source? obviously not; it has always been most extreme in universities. so where in universities did it begin? did it begin in math, or the hard sciences, or engineering, and spread from there to the humanities and social sciences? those are amusing images, but no, obviously it began in the humanities and social sciences.

why there? and why then? what happened in the humanities and social sciences in the 1980s?

a successful theory of the origin of political correctness has to be able to explain why it didn't happen earlier. why didn't it happen during the protest movements of the 1960s, for example? they were concerned with much the same issues. [1]

the reason the student protests of the 1960s didn't lead to political correctness was precisely that — they were student movements. they didn't have any real power. the students may have been talking a lot about women's liberation and black power, but it was not what they were being taught in their classes. not yet.

but in the early 1970s the student protestors of the 1960s began to finish their dissertations and get hired as professors. at first they were neither powerful nor numerous. but as more of their peers joined

them and the previous generation of professors started to retire, they gradually became both.

the reason political correctness began in the humanities and social sciences was that these fields offered more scope for the injection of politics. a 1960s radical who got a job as a physics professor could still attend protests, but his political beliefs wouldn't affect his work. whereas research in sociology and modern literature can be made as political as you like. [2]

i saw political correctness arise. when i started college in 1982 it was not yet a thing. female students might object if someone said something they considered sexist, but no one was getting *reported* for it. it was still not a thing when i started grad school in 1986. it was definitely a thing in 1988 though, and by the early 1990s it seemed to pervade campus life.

what happened? how did protest become punishment? why were the late 1980s the point at which protests against male chauvinism (as it used to be called) morphed into formal complaints to university authorities about sexism? basically, the 1960s radicals got tenure. they became the establishment they'd protested against two decades before. now they were in a position not just to speak out about their ideas, but to enforce them.

a new set of moral rules to enforce was exciting news to a certain kind of student. what made it particularly exciting was that they were allowed to attack professors. i remember noticing that aspect of political correctness at the time. it wasn't simply a grass-roots student movement. it was faculty members encouraging students to attack other faculty members. in that respect it was like the cultural revolution. that wasn't a grass-roots movement either; that was mao unleashing the younger generation on his political opponents. and in fact when roderick macfarquhar started teaching a class on the cultural revolution at harvard in the late 1980s, many saw it as a comment on current events. i don't know if it actually was, but people thought it was, and that means the similarities were obvious. [3]

college students larp. it's their nature. it's usually harmless. but larping morality turned out to be a poisonous combination. the result was a kind of moral etiquette, superficial but very complicated. imagine having to explain to a well-meaning visitor from another planet why using the phrase "people of color" is considered particularly enlightened, but saying "colored people" gets you fired. and why exactly one isn't supposed to use the word "negro" now, even though martin luther king used it constantly in his speeches. there are no underlying principles. you'd just have to give him a long list of rules to memorize. [4]

the danger of these rules was not just that they created land mines for the unwary, but that their elaborateness made them an effective substitute for virtue. whenever a society has a concept of heresy and orthodoxy, orthodoxy becomes a substitute for virtue. you can be the worst person in the world, but as long as you're orthodox you're better than everyone who isn't. this makes orthodoxy very attractive to bad people.

but for it to work as a substitute for virtue, orthodoxy must be difficult. if all you have to do to be orthodox is wear some garment or avoid saying some word, everyone knows to do it, and the only way to seem more virtuous than other people is to actually be virtuous. the shallow, complicated, and frequently changing rules of political correctness made it the perfect substitute for actual virtue. and the result was a world in which good people who weren't up to date on current moral fashions were brought down by people whose characters would make you recoil in horror if you could see them.

one big contributing factor in the rise of political correctness was the lack of other things to be morally pure about. previous generations of prigs had been prigs mostly about religion and sex. but among the cultural elite these were the deadest of dead letters by the 1980s; if you were religious, or a virgin, this

was something you tended to conceal rather than advertise. so the sort of people who enjoy being moral enforcers had become starved of things to enforce. a new set of rules was just what they'd been waiting for.

curiously enough, the tolerant side of the 1960s left helped create the conditions in which the intolerant side prevailed. the relaxed social rules advocated by the old, easy-going hippy left became the dominant ones, at least among the elite, and this left nothing for the naturally intolerant to be intolerant about.

another possibly contributing factor was the fall of the soviet empire. marxism had been a popular focus of moral purity on the left before political correctness emerged as a competitor, but the pro-democracy movements in eastern bloc countries took most of the shine off it. especially the fall of the berlin wall in 1989. you couldn't be on the side of the stasi. i remember looking at the moribund soviet studies section of a used bookshop in cambridge in the late 1980s and thinking "what will those people go on about now?" as it turned out the answer was right under my nose.

one thing i noticed at the time about the first phase of political correctness was that it was more popular with women than men. as many writers (perhaps most eloquently george orwell) have observed, women seem more attracted than men to the idea of being moral enforcers. but there was another more specific reason women tended to be the enforcers of political correctness. there was at this time a great backlash against sexual harassment; the mid 1980s were the point when the definition of sexual harassment was expanded from explicit sexual advances to creating a "hostile environment." within universities the classic form of accusation was for a (female) student to say that a professor made her "feel uncomfortable." but the vagueness of this accusation allowed the radius of forbidden behavior to expand to include talking about heterodox ideas. those make people uncomfortable too. [5]

was it sexist to propose that darwin's greater male variability hypothesis might explain some variation in human performance? sexist enough to get larry summers pushed out as president of harvard, apparently. one woman who heard the talk in which he mentioned this idea said it made her feel "physically ill" and that she had to leave halfway through. if the test of a hostile environment is how it makes people feel, this certainly sounds like one. and yet it does seem plausible that greater male variability explains some of the variation in human performance. so which should prevail, comfort or truth? surely if truth should prevail anywhere, it should be in universities; that's supposed to be their specialty; but for decades starting in the late 1980s the politically correct tried to pretend this conflict didn't exist. [6]

political correctness seemed to burn out in the second half of the 1990s. one reason, perhaps the main reason, was that it literally became a joke. it offered rich material for comedians, who performed their usual disinfectant action upon it. humor is one of the most powerful weapons against priggishness of any sort, because prigs, being humorless, can't respond in kind. humor was what defeated victorian prudishness, and by 2000 it seemed to have done the same thing to political correctness.

unfortunately this was an illusion. within universities the embers of political correctness were still glowing brightly. after all, the forces that created it were still there. the professors who started it were now becoming deans and department heads. and in addition to their departments there were now a bunch of new ones explicitly focused on social justice. students were still hungry for things to be morally pure about. and there had been an explosion in the number of university administrators, many of whose jobs involved enforcing various forms of political correctness.

in the early 2010s the embers of political correctness burst into flame anew. there were several differences between this new phase and the original one. it was more virulent. it spread further into the real

world, although it still burned hottest within universities. and it was concerned with a wider variety of sins. in the first phase of political correctness there were really only three things people got accused of: sexism, racism, and homophobia (which at the time was a neologism invented for the purpose). but between then and 2010 a lot of people had spent a lot of time trying to invent new kinds of -isms and -phobias and seeing which could be made to stick.

the second phase was, in multiple senses, political correctness metastasized. why did it happen when it did? my guess is that it was due to the rise of social media, particularly tumblr and twitter, because one of the most distinctive features of the second wave of political correctness was the *cancel mob*: a mob of angry people uniting on social media to get someone ostracized or fired. indeed this second wave of political correctness was originally called "cancel culture"; it didn't start to be called "wokeness" till the 2020s.

one aspect of social media that surprised almost everyone at first was the popularity of outrage. users seemed to *like* being outraged. we're so used to this idea now that we take it for granted, but really it's pretty strange. being outraged is not a pleasant feeling. you wouldn't expect people to seek it out. but they do. and above all, they want to share it. i happened to be running a forum from 2007 to 2014, so i can actually quantify how much they want to share it: our users were about three times more likely to upvote something if it outraged them.

this tilt toward outrage wasn't due to wokeness. it's an inherent feature of social media, or at least this generation of it. but it did make social media the perfect mechanism for fanning the flames of wokeness. [7]

it wasn't just public social networks that drove the rise of wokeness though. group chat apps were also critical, especially in the final step, cancellation. imagine if a group of employees trying to get someone fired had to do it using only email. it would be hard to organize a mob. but once you have group chat, mobs form naturally.

another contributing factor in this second wave of political correctness was the dramatic increase in the polarization of the press. in the print era, newspapers were constrained to be, or at least seem, politically neutral. the department stores that ran ads in the new york times wanted to reach everyone in the region, both liberal and conservative, so the times had to serve both. but the times didn't regard this neutrality as something forced upon them. they embraced it as their duty as a *paper of record* — as one of the big newspapers that aimed to be chronicles of their times, reporting every sufficiently important story from a neutral point of view.

when i grew up the papers of record seemed timeless, almost sacred institutions. papers like the new york times and washington post had immense prestige, partly because other sources of news were limited, but also because they did make some effort to be neutral.

unfortunately it turned out that the paper of record was mostly an artifact of the constraints imposed by print. [8] when your market was determined by geography, you had to be neutral. but publishing online enabled — in fact probably forced — newspapers to switch to serving markets defined by ideology instead of geography. most that remained in business fell in the direction they'd already been leaning: left. on october 11, 2020 the new york times announced that "the paper is in the midst of an evolution from the stodgy paper of record into a juicy collection of great narratives." [9] meanwhile journalists, of a sort, had arisen to serve the right as well. and so journalism, which in the previous era had been one of the great centralizing forces, now became one of the great polarizing ones.

the rise of social media and the increasing polarization of journalism reinforced one another. in fact there arose a new variety of journalism involving a loop through social media. someone would say something controversial on social media. within hours it would become a news story. outraged readers would then post links to the story on social media, driving further arguments online. it was the cheapest source of clicks imaginable. you didn't have to maintain overseas news bureaus or pay for month-long investigations. all you had to do was watch twitter for controversial remarks and repost them on your site, with some additional comments to inflame readers further.

for the press there was money in wokeness. but they weren't the only ones. that was one of the biggest differences between the two waves of political correctness: the first was driven almost entirely by amateurs, but the second was often driven by professionals. for some it was their whole job. by 2010 a new class of administrators had arisen whose job was basically to enforce wokeness. they played a role similar to that of the political commissars who got attached to military and industrial organizations in the ussr: they weren't directly in the flow of the organization's work, but watched from the side to ensure that nothing improper happened in the doing of it. these new administrators could often be recognized by the word "inclusion" in their titles. within institutions this was the preferred euphemism for wokeness; a new list of banned words, for example, would usually be called an "inclusive language guide." [10]

this new class of bureaucrats pursued a woke agenda as if their jobs depended on it, because they did. if you hire people to keep watch for a particular type of problem, they're going to find it, because otherwise there's no justification for their existence. [11] but these bureaucrats also represented a second and possibly even greater danger. many were involved in hiring, and when possible they tried to ensure their employers hired only people who shared their political beliefs. the most egregious cases were the new "dei statements" that some universities started to require from faculty candidates, proving their commitment to wokeness. some universities used these statements as the initial filter and only even considered candidates who scored high enough on them. you're not hiring einstein that way; imagine what you get instead.

another factor in the rise of wokeness was the black lives matter movement, which started in 2013 when a white man was acquitted after killing a black teenager in florida. but this didn't launch wokeness; it was well underway by 2013.

similarly for the me too movement, which took off in 2017 after the first news stories about harvey weinstein's history of raping women. it accelerated wokeness, but didn't play the same role in launching it that the 80s version did in launching political correctness.

the election of donald trump in 2016 also accelerated wokeness, particularly in the press, where outrage now meant traffic. trump made the new york times a lot of money: headlines during his first administration mentioned his name at about four times the rate of previous presidents.

in 2020 we saw the biggest accelerant of all, after a white police officer asphyxiated a black suspect on video. at this point the metaphorical fire became a literal one, as violent protests broke out across america. but in retrospect this turned out to be peak woke, or close to it. by every measure i've seen, wokeness peaked in 2020 or 2021.

wokeness is sometimes described as a mind-virus. what makes it viral is that it defines new types of impropriety. most people are afraid of impropriety; they're never exactly sure what the social rules are or which ones they might be breaking. especially if the rules change rapidly. and since most people already worry that they might be breaking rules they don't know about, if you tell them they're breaking

a rule, their default reaction is to believe you. especially if multiple people tell them. which in turn is a recipe for exponential growth. zealots invent some new impropriety to avoid. the first people to adopt it are fellow zealots, eager for new ways to signal their virtue. if there are enough of these, the initial group of zealots is followed by a much larger group, motivated by fear. they're not trying to signal virtue; they're just trying to avoid getting in trouble. at this point the new impropriety is now firmly established. plus its success has increased the rate of change in social rules, which, remember, is one of the reasons people are nervous about which rules they might be breaking. so the cycle accelerates. [12]

what's true of individuals is even more true of organizations. especially organizations without a powerful leader. such organizations do everything based on "best practices." there's no higher authority; if some new "best practice" achieves critical mass, they *must* adopt it. and in this case the organization can't do what it usually does when it's uncertain: delay. it might be committing improprieties right now! so it's surprisingly easy for a small group of zealots to capture this type of organization by describing new improprieties it might be guilty of. [13]

how does this kind of cycle ever end? eventually it leads to disaster, and people start to say enough is enough. the excesses of 2020 made a lot of people say that.

since then wokeness has been in gradual but continual retreat. corporate ceos, starting with brian armstrong, have openly rejected it. universities, led by the university of chicago and mit, have explicitly confirmed their commitment to free speech. twitter, which was arguably the hub of wokeness, was bought by elon musk in order to neutralize it, and he seems to have succeeded — and not, incidentally, by censoring left-wing users the way twitter used to censor right-wing ones, but without censoring either. [14] consumers have emphatically rejected brands that ventured too far into wokeness. the bud light brand may have been permanently damaged by it. i'm not going to claim trump's second victory in 2024 was a referendum on wokeness; i think he won, as presidential candidates always do, because he was more charismatic; but voters' disgust with wokeness must have helped.

so what do we do now? wokeness is already in retreat. obviously we should help it along. what's the best way to do that? and more importantly, how do we avoid a third outbreak? after all, it seemed to be dead once, but came back worse than ever.

in fact there's an even more ambitious goal: is there a way to prevent any similar outbreak of aggressively performative moralism in the future — not just a third outbreak of political correctness, but the next thing like it? because there will be a next thing. prigs are prigs by nature. they need rules to obey and enforce, and now that darwin has cut off their traditional supply of rules, they're constantly hungry for new ones. all they need is someone to meet them halfway by defining a new way to be morally pure, and we'll see the same phenomenon again.

let's start with the easier problem. is there a simple, principled way to deal with wokeness? i think there is: to use the customs we already have for dealing with religion. wokeness is effectively a religion, just with god replaced by protected classes. it's not even the first religion of this kind; marxism had a similar form, with god replaced by the masses. [15] and we already have well-established customs for dealing with religion within organizations. you can express your own religious identity and explain your beliefs, but you can't call your coworkers infidels if they disagree, or try to ban them from saying things that contradict its doctrines, or insist that the organization adopt yours as its official religion.

if we're not sure what to do about any particular manifestation of wokeness, imagine we were dealing with some other religion, like christianity. should we have people within organizations whose jobs are to enforce woke orthodoxy? no, because we wouldn't have people whose jobs were to enforce christian

orthodoxy. should we censor writers or scientists whose work contradicts woke doctrines? no, because we wouldn't do this to people whose work contradicted christian teachings. should job candidates be required to write dei statements? of course not; imagine an employer requiring proof of one's religious beliefs. should students and employees have to participate in woke indoctrination sessions in which they're required to answer questions about their beliefs to ensure compliance? no, because we wouldn't dream of catechizing people in this way about their religion. [16]

one shouldn't feel bad about not wanting to watch woke movies any more than one would feel bad about not wanting to listen to christian rock. in my twenties i drove across america several times, listening to local radio stations. occasionally i'd turn the dial and hear some new song. but the moment anyone mentioned jesus i'd turn the dial again. even the tiniest bit of being preached to was enough to make me lose interest.

but by the same token we should not automatically reject everything the woke believe. i'm not a christian, but i can see that many christian principles are good ones. it would be a mistake to discard them all just because one didn't share the religion that espoused them. it would be the sort of thing a religious zealot would do.

if we have genuine pluralism, i think we'll be safe from future outbreaks of woke intolerance. wokeness itself won't go away. there will for the foreseeable future continue to be pockets of woke zealots inventing new moral fashions. the key is not to let them treat their fashions as normative. they can change what their coreligionists are allowed to say every few months if they like, but they mustn't be allowed to change what we're allowed to say. [17]

the more general problem — how to prevent similar outbreaks of aggressively performative moralism — is of course harder. here we're up against human nature. there will always be prigs. and in particular there will always be the enforcers among them, the aggressively conventional-minded. these people are born that way. every society has them. so the best we can do is to keep them bottled up.

the aggressively conventional-minded aren't always on the rampage. usually they just enforce whatever random rules are nearest to hand. they only become dangerous when some new ideology gets a lot of them pointed in the same direction at once. that's what happened during the cultural revolution, and to a lesser extent (thank god) in the two waves of political correctness we've experienced.

we can't get rid of the aggressively conventional-minded. [18] and we couldn't prevent people from creating new ideologies that appealed to them even if we wanted to. so if we want to keep them bottled up, we have to do it one step downstream. fortunately when the aggressively conventional-minded go on the rampage they always do one thing that gives them away: they define new heresies to punish people for. so the best way to protect ourselves from future outbreaks of things like wokeness is to have powerful antibodies against the concept of heresy.

we should have a conscious bias against defining new forms of heresy. whenever anyone tries to ban saying something that we'd previously been able to say, our initial assumption should be that they're wrong. only our initial assumption of course. if they can prove we should stop saying it, then we should. but the burden of proof is on them. in liberal democracies, people trying to prevent something from being said will usually claim they're not merely engaging in censorship, but trying to prevent some form of "harm". and maybe they're right. but once again, the burden of proof is on them. it's not enough to claim harm; they have to prove it.

as long as the aggressively conventional-minded continue to give themselves away by banning heresies, we'll always be able to notice when they become aligned behind some new ideology. and if we always

fight back at that point, with any luck we can stop them in their tracks.

the number of true things we can't say should not increase. if it does, something is wrong.

---

notes

0 this was not the original meaning of "woke," but it's rarely used in the original sense now. now the pejorative sense is the dominant one.

1 why did 1960s radicals focus on the causes they did? one of the people who reviewed drafts of this essay explained this so well that i asked if i could quote him:

> the middle-class student protestors of the new left rejected the socialist/marxist left as unhip. they were interested in sexier forms of oppression uncovered by cultural analysis (marcuse) and abstruse "theory". labor politics became stodgy and old-fashioned. this took a couple generations to work through. the woke ideology's conspicuous lack of interest in the working class is the tell-tale sign. such fragments as are, er, left of the old left are anti-woke, and meanwhile the actual working class shifted to the populist right and gave us trump. trump and wokeness are cousins.
>
> the middle-class origins of wokeness smoothed its way through the institutions because it had no interest in "seizing the means of production" (how quaint such phrases seem now), which would quickly have run up against hard state and corporate power. the fact that wokeness only expressed interest in other kinds of class (race, sex, etc) signalled compromise with existing power: give us power within your system and we'll bestow the resource we control — moral rectitude — upon you. as an ideological stalking horse for gaining control over discourse and institutions, this succeeded where a more ambitious revolutionary program would not have.

2 it helped that the humanities and social sciences also included some of the biggest and easiest undergrad majors. if a political movement had to start with physics students, it could never get off the ground; there would be too few of them, and they wouldn't have the time to spare.

at the top universities these majors are not as big as they used to be, though. a 2022 survey found that only 7% of harvard undergrads plan to major in the humanities, vs nearly 30% during the 1970s. i expect wokeness is at least part of the reason; when undergrads consider majoring in english, it's presumably because they love the written word and not because they want to listen to lectures about racism.

3 the puppet-master-and-puppet character of political correctness became clearly visible when a bakery near oberlin college was falsely accused of race discrimination in 2016. in the subsequent civil trial, lawyers for the bakery produced a text message from oberlin dean of students meredith raimondo that read "i'd say unleash the students if i wasn't convinced this needs to be put behind us."

4 the woke sometimes claim that wokeness is simply treating people with respect. but if it were, that would be the only rule you'd have to remember, and this is comically far from being the case. my younger son likes to imitate voices, and at one point when he was about seven i had to explain which accents it was currently safe to imitate publicly and which not. it took about ten minutes, and i still hadn't covered all the cases.

5 in 1986 the supreme court ruled that creating a hostile work environment could constitute sex discrimination, which in turn affected universities via title ix. the court specified that the test of a hostile environment was whether it would bother a reasonable person, but since for a professor merely being the subject of a sexual harassment complaint would be a disaster whether the complainant was reasonable or not, in practice any joke or remark remotely connected with sex was now effectively forbidden. which meant we'd now come full circle to victorian codes of behavior, when there was a large class of things that might not be said "with ladies present."

6 much as they tried to pretend there was no conflict between diversity and quality. but you can't simultaneously optimize for two things that aren't identical. what diversity actually means, judging from the way the term is used, is proportional representation, and unless you're selecting a group whose purpose is to be representative, like poll respondents, optimizing for proportional representation has to come at the expense of quality. this is not because of anything about representation; it's the nature of optimization; optimizing for x has to come at the expense of y unless x and y are identical.

7 maybe societies will eventually develop antibodies to viral outrage. maybe we were just the first to be exposed to it, so it tore through us like an epidemic through a previously isolated population. i'm fairly confident that it would be possible to create new social media apps that were less driven by outrage, and an app of this type would have a good chance of stealing users from existing ones, because the smartest people would tend to migrate to it.

8 i say "mostly" because i have hopes that journalistic neutrality will return in some form. there is some market for unbiased news, and while it may be small, it's valuable. the rich and powerful want to know what's really going on; that's how they became rich and powerful.

9 the times made this momentous announcement very informally, in passing in the middle of an article about a times reporter who'd been criticized for inaccuracy. it's quite possible no senior editor even approved it. but it's somehow appropriate that this particular universe ended with a whimper rather than a bang.

10 as the acronym dei goes out of fashion, many of these bureaucrats will try to go underground by changing their titles. it looks like "belonging" will be a popular option.

11 if you've ever wondered why our legal system includes protections like the separation of prosecutor, judge, and jury, the right to examine evidence and cross-examine witnesses, and the right to be represented by legal counsel, the de facto parallel legal system established by title ix makes that all too clear.

12 the invention of new improprieties is most visible in the rapid evolution of woke nomenclature. this is particularly annoying to me as a writer, because the new names are always worse. any religious observance has to be inconvenient and slightly absurd; otherwise gentiles would do it too. so "slaves" becomes "enslaved individuals." but web search can show us the leading edge of moral growth in real time: if you search for "individuals experiencing slavery" you will as of this writing find five legit attempts to use the phrase, and you'll even find two for "individuals experiencing enslavement."

13 organizations that do dubious things are particularly concerned with propriety, which is how you end up with absurdities like tobacco and oil companies having higher esg ratings than tesla.

14 elon did something else that tilted twitter rightward though: he gave more visibility to paying users. paying users lean right on average, because people on the far left dislike elon and don't want to give him money. elon presumably knew this would happen. on the other hand, the people on the far left have only themselves to blame; they could tilt twitter back to the left tomorrow if they wanted to.

15 it even, as james lindsay and peter boghossian pointed out, has a concept of original sin: privilege. which means unlike christianity's egalitarian version, people have varying degrees of it. an able-bodied straight white american male is born with such a load of sin that only by the most abject repentance can he be saved.

wokeness also shares something rather funny with many actual versions of christianity: like god, the people for whose sake wokeness purports to act are often revolted by the things done in their name.

16 there is one exception to most of these rules: actual religious organizations. it's reasonable for them to insist on orthodoxy. but they in turn should declare that they're religious organizations. it's rightly considered shady when something that appears to be an ordinary business or publication turns out to be a religious organization.

17 i don't want to give the impression that it will be simple to roll back wokeness. there will be places where the fight inevitably gets messy — particularly within universities, which everyone has to share, yet which are currently the most pervaded by wokeness of any institutions.

18 you can however get rid of aggressively conventional-minded people within an organization, and in many if not most organizations this would be an excellent idea. even a handful of them can do a lot of damage. i bet you'd feel a noticeable improvement going from a handful to none.

thanks to sam altman, ben miller, daniel gackle, robin hanson, jessica livingston, greg lukianoff, harj taggar, garry tan, and tim urban for reading drafts of this.

13

the brand age

march 2026
paulgraham.com/brandage.html

in the early 1970s, disaster struck the swiss watch industry. now people call it the quartz crisis, but in fact it was a compound of three separate disasters that all happened at about the same time.

the first was competition from japan. the swiss had been watching the japanese in the rear view mirror all through the 1960s, and they'd been improving at an alarming rate. but even so the swiss were surprised in 1968 when the japanese swept all the top spots for mechanical watches at the geneva observatory trials.

the swiss knew what was coming. for years the japanese had been able to make cheaper watches. now they could make better ones too.

to make matters worse, swiss watches were about to become much more expensive. the bretton woods agreement, which since 1945 had fixed the exchange rates of most of the world's currencies, had set the swiss franc at an artificially low rate of .228 usd. when bretton woods collapsed in 1973, the franc shot upward. by 1978 it reached .625 usd, meaning swiss watches were now 2.7 times as expensive for americans to buy. [1]

the combined effect of foreign competition and the loss of their protective exchange rate would have decimated the swiss watch industry even if it hadn't been for quartz movements. but quartz movements were the final blow. now the whole game they'd been trying to win at became irrelevant. something that had been expensive — knowing the exact time — was now a commodity.

between the early 1970s and the early 1980s, unit sales of swiss watches fell by almost two thirds. most swiss watchmakers became insolvent or close to it and were sold. but not all of them. a handful survived as independent companies. and the way they did it was by transforming themselves from precision instrument makers into luxury brands.

in the process the nature of the mechanical watch was also transformed. the most expensive watches have always cost a lot, but why they cost a lot and what buyers got in return have changed completely. in 1960 expensive watches cost a lot because they cost a lot to manufacture, and what the buyer got in return was the most accurate timekeeping device, for its size, that could be made. now they cost a lot because brands spend a lot on advertising and use tricks to limit supply, and what the buyer gets in return is an expensive status symbol.

that turns out to be a profitable business though. the swiss watch industry probably makes more now from selling brand than they would have if they were still selling engineering. and indeed, when you look at the graph of swiss watch sales by revenue, it tells a different story than the graph of unit sales. instead of falling off a cliff, the revenue numbers merely flatten out for a while, and then take off like a rocket in the late 1980s as the surviving watchmakers come to terms with their new destiny.

it took the watchmakers about 20 years to figure out the new rules of the game. and it's interesting to watch them do it, because the completeness of their transformation makes it the perfect case study in one of the most powerful forces of our era: brand.

brand is what's left when the substantive differences between products disappear. but making the substantive differences between products disappear is what technology naturally tends to do. so what happened to the swiss watch industry is not merely an interesting outlier. it's very much a story of our times.

jaeger-lecoultre's web site says that one of their current collections "takes its inspiration from the classic designs of the golden age of watchmaking." in saying this they're implicitly saying something that

present-day watchmakers all know but rarely come so close to saying outright: whatever age we're in now, it's not the golden age.

the golden age was from 1945 to 1970 — from the point where the watch industry emerged from the chaos of war with the swiss on top till the triple cataclysm that struck it starting in the late 60s. there were two things watchmakers sought above all in the golden age: thinness and accuracy. and indeed this was arguably the essential tradeoff in watchmaking. a watch is something you carry with you to tell you the time. so there are two fundamental ways to improve it: to make it easier to carry with you and to make it better at telling the time.

obviously accuracy is valuable, but in the golden age thinness was if anything more valuable. even in the days of pocket watches the best watchmakers tried to make their watches as thin as they could. cheap, thick pocket watches were derided as "turnips." but thinness took on a new urgency when men's watches moved onto their wrists during world war i. and since thinness was more difficult to achieve than accuracy, it was this quality that tended to distinguish the more expensive watches of the golden age.

there is one other thing watchmakers have pursued in some eras: telling more than the time in the usual way. telling you the phase of the moon, for example, or telling the time with sound. in the industry the term for these things is "complications." they were popular in the nineteenth century and they're popular again now, but except for one pragmatic complication (showing the date), they were a sideshow in the golden age. in the golden age, as always in golden ages, the top watchmakers focused on the essential tradeoff. and, as always in golden ages, they did it beautifully. the best watches of the golden age have a quiet perfection that has never been equalled since. and for reasons i'm about to explain, probably never will be.

the three most prestigious brands of the golden age were the so-called "holy trinity" of patek philippe, vacheron constantin, and audemars piguet. their prestige was mostly deserved; they had earned it by the exceptional quality of their work. by the 1960s they stood on two legs, prestige and performance. and what they learned in the next two decades was that they had to put all their weight on the first leg, because they could no longer win at either of the two things watchmakers had historically striven to achieve. quartz movements were not only more accurate than any mechanical movement, but thinner too.

the holy trinity at least had another leg to stand on. most of the other well-known swiss watchmakers sold only performance. none of those companies survived intact.

omega showed what not to do. omega were the nerds of swiss watchmakers. they made wonderfully accurate watches, but they would have been ambivalent, at best, about the idea of being a luxury brand. when the japanese got as good as the swiss at making accurate movements, omega responded in the omega way: make even more accurate movements. they introduced a new movement in 1968 that ran at a 45% higher frequency. in theory this should have made it more accurate, but the new movement was so fragile that it destroyed their reputation for reliability. they even tried to make a better quartz movement, but there was nothing down that road but a race to the bottom. by 1981 they were insolvent and were taken over by their creditors.

patek philippe took the opposite approach. while omega was redesigning their movements, patek was redesigning their cases. or more precisely, designing their cases, because until then they hadn't.

this is probably the point to mention what a strange beast the swiss watch industry was in those days. it was a kind of capitalism that's hard to imagine today, and even then could only have been made to

work in a country like switzerland — a network of small, specialized companies locked into place by regulation. the companies that we for convenience have been calling watchmakers were merely the consumer-facing edge of this network. the holy trinity didn't design their own cases, or even their own movements most of the time.

in 1968 (that year again) patek philippe launched a new watch that shifted the center of gravity of case design. this time they'd taken their own designs to the casemakers and said "this is what you're going to make for us." the result was a striking new model called the golden ellipse. somewhat confusingly, because it wasn't elliptical. the new case was more of what ui designers would call a round rect: a rectangle with rounded corners. and this new family of watches was quite successful. but it was more than that: it was the pattern for the future. [2]

how could merely designing a distinctive case be so important? because it turned the entire watch into an expression of brand.

the trouble with the best watches of the golden age, from the point of view of someone who wanted to impress people with the brand of watch he was wearing, was that no one could tell what brand of watch you were wearing. until you got within a few inches of them, the watches of all the top makers looked the same. that's the thing about minimalism: there tends to be just one answer. plus the watches of the golden age were small by present standards. watchmakers had spent centuries working to make them smaller, and by 1960 they'd gotten very good at it. so the only thing distinguishing one top brand from another was the name printed on the dial, and dials were so small that these names were tiny. the manufacturers' names on the holy trinity's golden age watches are between half and three quarters of a millimeter high. by taking over the case, patek expanded the size of the brand from 8 square millimeters to 800.

why did they suddenly decide to make their brand shout, after a century of whispering? because they knew they weren't going to beat the japanese on performance. from now on they'd have to depend more on brand.

there's a cost to doing this, which we can see even in this early example of case-as-brand. golden ellipses are not bad looking. they must have looked even cooler in the 1970s, when designers were turning everything into round rects. but the golden ellipse was not an evolutionary step forward in case design. watches didn't all become round rects. watchmakers had already discovered the optimal shape for the case of something that describes a circle as it rotates.

they had also discovered the optimal shape for the crown, the knob on the side of a watch that you turn to wind it. but to emphasize the distinctive profile of the ellipse, patek made the crown too small, with the result that they're distractingly hard to wind. [3]

so even in this early example we see an important point about the relationship between brand and design. branding isn't merely orthogonal to good design, but opposed to it. branding by definition has to be distinctive. but good design, like math or science, seeks the right answer, and right answers tend to converge.

branding is centrifugal; design is centripetal.

there is some wiggle room here of course. design doesn't have as sharply defined right answers as math, especially design meant for a human audience. so it's not necessarily bad design to do something distinctive if you have honest motives. but you can't evade the fundamental conflict between branding and design, any more than you can evade gravity.

indeed, the conflict between branding and design is so fundamental that it extends far beyond things we call design. we see it even in religion. if you want the adherents of a religion to have customs that set them apart from everyone else, you can't make them do things that are convenient or reasonable, or other people would do them too. if you want to set your adherents apart, you have to make them do things that are inconvenient and unreasonable.

it's the same if you want to set your designs apart. if you choose good options, other people will choose them too.

there are only two ways to combine branding and good design. you can do it when the space of possibilities is enormously large, as it is in painting, for example. leonardo could paint as well as he possibly could and yet also paint in a style that was distinctively his. if there had been a million painters as good as bellini and leonardo this would have been harder to do, but since there were more like ten they didn't bump up against one another much. [4]

the other situation when branding and good design can be combined is when the space of possibilities is comparatively unexplored. if you're the first to arrive in some new territory, you can both find the right answer and claim it as uniquely yours. at least at first; if you've really found the right answer, everyone else's designs will inevitably converge on yours, and your brand advantage will erode over time.

since the space of watch design is neither unexplored nor enormously large, branding can only be achieved at the expense of good design. and in fact if you wanted one sentence to describe the current age of watchmaking, that one would do pretty well.

patek philippe didn't know for sure that making visibly branded watches would work. it was not even their only strategy, at the time. they were finding their way. but it was the strategy that did work, at least as measured by revenues.

for it to work the customers had to meet them halfway. patek knew that not all their customers were buying their watches for the performance they delivered — for their accuracy and thinness. they knew that at least some customers were buying them because they were expensive. but it was unclear how many, or how far they could be pushed.

to encourage them, patek did something that none of the holy trinity had done much of before: brand advertising. and what they talked about was how expensive their watches were. a 1968 patek ad explained "why you are well advised to invest perhaps half a month's income" in an ellipse. "like every patek philippe," the ad continued, "this thin model is entirely finished by hand. since a patek philippe is the costliest watch to make, production is severely limited: only 43 watches are signed out each day for delivery to prominent jewelers throughout the world." [5]

you can tell this is an early ad because they still mention thinness. but there is no mention of accuracy. presumably patek felt that battle was already lost.

the next move was made by audemars piguet, who in 1970 commissioned the renowned designer gérald genta to design their own iconic watch, this one, daringly, in steel. the result, launched in 1972, was the royal oak. and audemars piguet's ads (for they too now started doing brand advertising) emphasized its high cost even more dramatically. "introducing steel at the price of gold," one began. "you're looking at the costliest stainless steel watch in the world — the audemars piguet 'royal oak'. what makes it even more precious than gold is the time that went into building it, by a vanishing breed of master watchmakers." at the bottom of the ad they turn the traditional formula on its head and describe their watches as being "priced from $35,000 and down."

the royal oak was also a step forward in surface area devoted to brand. the golden ellipse had turned the watch face into an expression of brand, but it used ordinary straps and bracelets. in the royal oak, the watch face was integrated with a metal bracelet that continued its design all the way around the wrist. when it said "you're looking at the costliest stainless steel watch in the world," it said it with every square millimeter of surface area.

would customers buy this new approach? the initial results were moderately encouraging. the holy trinity's sales didn't take off, but they didn't go down to zero either. there were at least some people out there responding to the new message. perhaps if they kept at it the number would grow.

so they did. encouraged by the success of the royal oak, patek philippe commissioned gérald genta in 1974 to design a similar watch for them. the design of the royal oak had been inspired by a ship's porthole, so the design of this new watch would be inspired by... a ship's porthole. it was called the nautilus, and it launched at the basel watch fair in 1976.

in the nautilus we really see the incompatibility of branding and design. it was huge. the most expensive men's watches at the peak of the golden age were typically 32 or 33 millimeters in diameter. the nautilus was 42 millimeters. and as well as being huge it had gratuitous knobs on either side of the face, like a pair of ears. but you could recognize one from across the room.

of all the watches patek makes now, the nautilus is the most sought after. it's perfectly aligned with what present-day buyers want — basically, the loudest possible expression of brand. but in 1976 it was ahead of its time. in 1976 it was still a little too much.

the watch that finally turned patek's fortunes around was another iconic design, the hobnail calatrava. the hobnail calatravas were so called because they were decorated with tiny pyramid-shaped spikes. that was enough to make them look distinctive. but except for the hobnails they were basically golden age dress watches.

the hobnail calatrava was apparently the brainchild of rené bittel, the head of patek philippe's ad agency. it was not a new design. many watchmakers had decorated their cases with hobnails over the years, and there had been a patek model with them since 1968. but in 1984 bittel told patek president philippe stern, in effect: make this your standard design, and i'll create an ad campaign to identify it in people's heads with your brand. [6]

it worked spectacularly well. the resulting watch, the 3919, is known as the "banker's watch" because it became so popular among investment bankers in new york in the 80s and 90s. up to this point patek had been hedging their bets, making quartz watches as well, and arguing defensively in their ads that quartz watches in fancy cases were almost as laborious to make as mechanical ones. but the ibankers bought the full mechanical story. they didn't even need self-winding mechanical watches; the 3919 was hand-wound. so be it. patek stopped talking about quartz movements. and their sales, which had been flat since the early 70s, were by 1987 on a clear upward trajectory.

it's hard to say for sure whether the critical ingredient was bittel's skill at advertising or a receptive audience, but as someone who knew these investment bankers, i'd lean toward the audience. these were the people for whom the term "yuppy" was coined. living expensively was one of the things they were best known for. if anyone was going to adopt a new way to display wealth, it would be them. whereas if bittel had sent the same message ten years earlier, there might have been no one to hear it.

whatever the cause, something happened in the second half of the 1980s, because that's when all the numbers finally start going up again. up till about 1985 it was still not clear what would happen with

mechanical watches. by 1990 it was. by 1990 the custom of using expensive, highly-branded, conspicuously mechanical watches as status symbols was firmly established. [7]

obsolete technologies don't usually get adopted as ways to display wealth. why did it happen with mechanical watches? because the wristwatch turns out to be the perfect vehicle for it. where better than right on your wrist, where everyone can see it? and more to the point, what better to do it with? you could wear a diamond ring or a gold chain, but those would have seemed socially dubious to investment bankers. they might have been barbarians, but they weren't mafia. whereas nothing could be more legit than a gold watch. the chairman of the company was still wearing one his wife gave him 20 years ago, before quartz watches were even a thing. if the increasing pressure to display wealth was going to emerge anywhere, this was the place. [8]

for men, at least. women never really went for the idea of wearing mechanical watches. most rich women are happy wearing a cartier tank with a quartz movement. why the difference? partly for the same reason that most buyers of steam engines are men. but the main reason is that expensive mechanical watches now serve as de facto jewelry for men, and women don't need de facto jewelry because they can wear actual jewelry.

it was critical, though, that mechanical watches were accurate *enough*. a new 3919 would have been off by no more than 5 seconds a day. that was nowhere near as good as quartz. even the cheapest mass market quartz watches were accurate to half a second a day, and the best ones were accurate to 3 seconds a year. but in practice you didn't need that kind of accuracy. if mechanical watches had only been accurate to a minute a day they couldn't have made the leap from keeping time to displaying wealth. it would have seemed too manifestly unluxurious to have a watch that always had the wrong time. but 5 seconds a day was close enough. [9]

this is an important point about the relationship between brand and quality. quality doesn't stop mattering when a product switches to something people buy for its brand. but the way it matters changes shape. it becomes a threshold. it no longer has to be so great that it sells the product; brand sells the product; but it does have to be good enough to maintain the brand's reputation. the brand must not break character.

it was a lucky thing for the watchmakers that yuppies arose just in time to save them. or maybe not so lucky. because the evolution of the market that yuppies represented has continued with a vengeance, and watchmakers have perforce been dragged along with it. if they don't make gigantic blingy watches for buyers in hong kong and dubai, someone else will. so that is what they now find themselves doing. and what began with a few comparatively subtle examples of the conflict between branding and design is now an all out war on design.

the present era of mechanical watchmaking doesn't yet have a name. but if we need one, it's obvious what it should be: the brand age. the golden age ran from 1945 to 1970, followed by the quartz crisis from 1970 to 1985. since 1985 we've been in the brand age.

this won't be the only brand age. indeed, it's not even the first; fine art has been in its own brand age since the establishment of the barr canon in the 1930s. and since we'll probably see more of this kind of thing, it would be worth taking some time to look at what a brand age is like.

how are things different now from the way they were in the golden age? the best way to answer that might be to imagine what someone from the golden age would notice if we brought him here in a time machine.

the first thing he'd notice, if he walked through a fancy shopping district, is that all the prominent watchmakers of the golden age seem to be doing better than ever. they're not only all still around, but most now have their own boutiques instead of depending on jewelers to sell their products as they used to back in the day.

in fact this is an illusion. only three watchmakers survived the dark days of the 70s and 80s as independent companies: patek philippe, audemars piguet, and rolex. all the rest are owned by six holding companies, which reinflated them as it became clear that mechanical watches would have a second life as luxury accessories for men. instead of separate companies they're now more like the brands that got rolled up into the big three american automakers: they're ways for their parent companies to target different segments of the market. so longines, for example, no longer competes with omega, because the company that owns them both has assigned it a lower tier of the market. [10]

there's a reason the vacheron constantin boutique looks so much like the iwc and jaeger-lecoultre boutiques, and for that matter the montblanc and cartier boutiques. they're all owned by the same company. it's similar with clothing brands, incidentally. when you walk through a town's fanciest shopping district, what seem to be the shops of lots of different brands are actually owned by a handful of conglomerates. that's one reason these districts seem so sterile; like suburbs built by a single developer, they have an unnatural lack of variety.

when our time traveler peered into the windows of these shops, the first thing he'd notice would be how large all the watches were. this would surprise him, because in the golden age, as indeed in all the preceding centuries, big meant cheap. an expensive golden age men's watch might have been 33 millimeters in diameter and 8 millimeters thick. an expensive watch today will be more like 42 millimeters in diameter and 10 millimeters thick — more than double the size. it would astonish our visitor to look through the windows of what were clearly very fancy shops and see what seemed to be cheap watches. [11]

we know how this happened. when watches switched from telling time to telling brand, they grew in size to be better at it. and not just in size, but in shape too. that's another thing our time traveler would notice: the surprising variety of strange case shapes and awkward protrusions that have been produced as the centrifugal tendency of branding played out. what, he'd wonder, is going on with the huge guards on the crowns of those panerais? what do people do with these watches that makes the crown need such protection? and why would a crown guard have a message engraved on it saying that it's a registered trademark? it's obvious to us what's going on here, but imagine how confusing it would be to someone from the golden age, when form followed function. [12]

as he puzzled over this strange assortment of bulky watches, he'd notice a further pattern. he'd realize that a surprisingly large number of them looked like a specific brand of bulky watch he was already familiar with.

i haven't talked about rolex so far, because rolex didn't have to do much to adapt to the new era. they already had one foot in the brand age during the golden age. early in their history they put a lot of effort into making their watches better, but they "stopped taking part in competitions in geneva and neuchâtel at the end of the 1950s," and from about 1960 "largely abandoned research into mechanical watchmaking." [13] the reason was not that they'd become lazy, but that they'd discovered they could make sales grow faster by marketing their watches as status symbols. so that became their focus during the 1960s, and by the time the quartz crisis hit ten years later, their customers were self-selected to be people who didn't care that much what was inside a watch, so long as it was recognizably a rolex.

and they were far ahead of other watchmakers in that department. they already had in the 1940s what we saw patek philippe and audemars piguet struggling to create in the 1970s and 80s: a case that immediately proclaimed the brand of the maker. the rolex look seems to have evolved organically, but once it did, they realized how important it was. in fact they pitched it as one of the features of their watches. a 1960s rolex ad says "you can recognize its classic shape, carved out of a block of solid gold, from the other end of the conference table."

indeed rolex was ahead of its time in both dimensions: their cases were not merely recognizable, but big too, at least by golden age standards. that was not the result of clever marketing, though. it was a byproduct of the founder hans wilsdorf's obsession with building waterproof watches.

as its name suggests, that was the raison d'etre of the rolex oyster. watches like the oyster were designed to be tough, like jeeps. in the golden age there were two poles of watch design. at one end were tool watches, which were thick, tough, and usually made of steel. at the other end were dress watches, which were thin, elegant, and usually made of gold. but rolex blurred the line between them. when they made thick, tough watches, they made them out of gold as well as steel. the result was a sort of luxury jeep. and if that phrase didn't ring a bell in your head, stop and think about it, because that is exactly what everyone is driving now. that's what suvs are, luxury jeeps. what happened to watches is the same thing that happened to cars. and indeed if our time traveler turned and saw a porsche cayenne pass by and realized what it was — a huge, pseudo-offroad vehicle meant to recall the porsche 911 — he might have been even more shocked than he was by the watches he'd been looking at. [14]

if the time traveller walked into a patek philippe boutique and actually tried to buy a nautilus, he'd get the biggest shock of all. they wouldn't sell him one. because at patek he'd encounter the most extreme brand age phenomenon: artificial scarcity. you can't just buy a nautilus. you have to spend years proving your loyalty first by buying your way through multiple tiers of other models, and then spend years on a waiting list. [15]

obviously this strategy sells more watches. but it also supports retail prices by keeping watches off the secondary market. a company using artificial scarcity to drive sales can't allow too many of the scarce models to leak into the secondary market, or they stop being scarce. the ideal is the watch equivalent of carbon sequestration: for the people who buy their watches to keep them till they die.

to push the market toward this ideal, patek squeezes from both sides of the sale. they weed out flippers by making the path to the scarce models so costly in both time and money — so inconvenient and unreasonable — that only a genuine fan would endure it. the lower tier watches sell for below retail on the secondary market, because patek *doesn't* restrict their supply, so a would-be flipper should have to spend years making money-losing purchases before he could even get something he could flip at a profit. apparently some people still manage to beat this system though, so patek's countermeasures don't end there. they keep a vigilant eye on secondary sales to see who's selling their watches. auction listings usually include serial numbers, so those are easy to trace, but if necessary they'll rebuy their own watches on the secondary market to get the serial number and trace the leak. they buy hundreds a year. and when they catch someone selling watches they don't want them to, they don't just cut off that customer. if a retailer's customers are responsible for too many such leaks, they'll cut off the whole retailer. which naturally makes retailers eager to help them police buyers.

there will of course always be some leaks into the secondary market. even the most loyal customers die at a certain rate. and in fact it's critical for patek that the secondary market continue to exist, because it's one of the most valuable sources of information they have about the most important question they face: how fast to increase the supply of the top tier watches. their scarcity helps drive the purchases of

all the others, so those that do make it into the secondary market should always sell for above retail. and i'm sure patek leaves a large margin for error when increasing supply, because if secondary market prices for these watches get close to retail prices, you're getting close to a price collapse — which, since people now buy these watches as investments, would have the same disastrous cascading effect as the bursting of an asset bubble. it wouldn't just be like the bursting of an asset bubble. it would *be* the bursting of an asset bubble. that's the business an elite watchmaker is in now: carefully managing a sustained asset bubble. [16]

this is an instance of what i call the comb-over effect: when a series of individually small changes takes you from something that's a little bit off to something that's freakishly wrong. i'm sure patek didn't cook up this whole scheme in one shot; i'm sure it evolved gradually. but look at what a strange place we've ended up in. back in the golden age the way you bought a patek philippe was to go to a jeweler and give them money. now patek is policing buyers to maintain an asset bubble.

the most striking thing to me about the brand age is the sheer strangeness of it. the zombie watch brands that appear to be independent and even have their own retail stores, and yet are all owned by a few holding companies. the giant, awkwardly shaped watches that reverse 500 years of progress in making them smaller. the business model that requires a company to rebuy their own watches on the secondary market to catch rogue customers. the very concept of rogue customers. it's all so strange. and the reason it's strange is that there's no function for form to follow.

up to the end of the golden age, mechanical watches were necessary. you needed them to know the time. and that constraint gave both the watches and the watchmaking industry a meaningful shape. there were certainly some strange-looking watches made during the golden age. they weren't all beautifully minimal. but when golden age watchmakers made a strange-looking watch, they knew they were doing it. in fact they give the impression of having done it as a deliberate exercise, to avoid getting into a rut.

that's not why brand age watches look strange. brand age watches look strange because they have no practical function. their function is to express brand, and while that is certainly a constraint, it's not the clean kind of constraint that generates good things. the constraints imposed by brand ultimately depend on some of the worst features of human psychology. so when you have a world defined only by brand, it's going to be a weird, bad world.

well that was dark. is there some edifying lesson we can salvage from the wreckage?

one obvious lesson is to stay away from brand. indeed it's probably a good idea not just to avoid buying brand, but to avoid selling it too. sure, you might be able to make money this way — though i bet it's harder than it looks — but pushing people's brand buttons is just not a good problem to work on, and it's hard to do good work without a good problem.

the more subtle lesson is that fields have natural rhythms that are beyond the power of individuals to resist. fields have golden ages and not so golden ages, and you're much more likely to do good work in a field that's on the way up.

of course they don't call them golden ages as they're happening. "golden age" is a term people use later, after they're over. that doesn't mean that golden ages aren't real, but rather that their participants take them for granted at the time. they don't know how good they have it. but while it's usually a mistake to take one's good fortune for granted, it's not in this case. what a golden age feels like, at the time, is just that smart people are working hard on interesting problems and getting results. it would be overfitting to optimize for more than that.

in fact there's a single principle that will both save you from working on things like brand, and also automatically find golden ages for you. follow the problems.

the way to find golden ages is not to go looking for them. the way to find them — the way almost all their participants have found them historically — is by following interesting problems. if you're smart and ambitious and honest with yourself, there's no better guide than your taste in problems. go where interesting problems are, and you'll probably find that other smart and ambitious people have turned up there too. and later they'll look back on what you did together and call it a golden age.

---

notes

1 the bretton woods agreement didn't fix exchange rates between currencies directly. it fixed each relative to gold. obviously this also fixed them relative to one another.

2 the golden ellipse isn't quite a round rect, because the sides aren't quite flat. it's similar in shape to the superellipses popularized by piet hein in the early 1960s, and in fact that may be where they got the name. but mathematically it's not an actual superellipse. my guess is that patek's designer just experimented with french curves till he got something he liked. and to be fair it is a good shape.

3 it was ironic that patek philippe of all companies made this mistake, because adrien philippe was the inventor of the modern crown. but they must have realized what they'd done, because later ellipses have if anything excessively prominent crowns.

4 the high ratio of design space to practitioners in fine art has combined with the practical importance of attribution to give people the impression that painting in a distinctively leonardesque way is what makes leonardo good. the most dangerous problem faced by curators, art historians, and art dealers — the one that has the worst consequences if they get the wrong answer — is attribution. so inevitably they spend a lot of time thinking and talking about the features that distinguish the work of one artist from another. but those aren't what make artists good. what makes the line of a woman's cheek in a leonardo drawing good is how good it looks as the line of a cheek, not how little it looks like lines made by other artists.

because painting has such prestige, the myth that having a distinctive style (rather than painting well) is the defining quality of great artists has in turn given cover to a lot of bad design in adjacent fields. a brand that does something hideous to distinguish their products can say "like all great works of art, ours have a distinctive style," and people will buy it.

5 an ad that patek philippe ran in america in 1970 famously described a patek 3548 with a gold bracelet as a "$1700 trust fund." was it actually a good investment? in the very best case a dealer might pay you $20k now for one in unworn condition with its original box and papers. that's about a 4.5% rate of return, which is not absolutely terrible. but apparently the average rate of return on s&p 500 stocks over this period was more like 10%, if you reinvested all the dividends after paying taxes on them. the average rate of return would have been over 9% if you merely bought a lump of gold that hadn't been made into a watch. so, not surprisingly, the ad wasn't very good investment advice.

6 tania edwards, who ran us marketing for patek philippe in the 90s, said that bittel literally sketched the design of the 3919 on a piece of paper. this sounds odd to me, because the 3919 looked exactly like the existing 3520 with the addition of sub seconds (a small dial with a second hand above 6 o'clock). why would you sketch a design almost identical to an existing watch when you could just point to the existing watch and say "that, with sub seconds." what this story does show, though, is the degree to which people within patek felt their ad agency was responsible for the design of the 3919.

7 if i had to date the turning point for mechanical watches precisely, i'd say 1986. unit sales of swiss watches rebounded in 1985, but revenue didn't, which means what we're seeing is the boom in cheap quartz swatches. indeed, sales of mechanical watches must have been down if revenue was flat despite the sale of all those

swatches. whereas in 1986 revenue turns sharply upward even though unit sales only increase by a little, which implies a corresponding increase in sales of expensive mechanical watches.

8 there is of course another reason some people are into mechanical watches: because they're interested in old technology. and if you are genuinely interested in mechanical watches, there's good news. you don't have to wear a billboard on your wrist or pay a lot to own one. just buy golden age watches. they still keep good time, they're much more beautiful, and they cost a fraction of what new watches cost.

the key to buying a golden age watch is to find a good dealer, and the best way to recognize one is by how much they tell you about the watch. a bad dealer will just have a lot of fluff about the prestige of the brand and the sleek lines of the case. a good dealer will tell you the model number of the watch and movement, have lots of pictures, including some with the case back open, give you dimensions, disclose all damage and restoration, and tell you exactly how accurately the watch is running. good dealers tend to be watch nerds themselves, so they're into this kind of thing.

(there are a few independent watchmakers trying earnestly to make good mechanical watches now, but their efforts show how hard it is to do good work when the current is against you.)

9 oddly enough it might have helped that the 3919 was hand wound. if a watch runs for long enough, 5 seconds a day starts to add up. after three months a watch that gains 5 seconds a day will be 7 minutes fast. but with a hand wound watch you occasionally forget to wind it, and it runs down. and when you wind it again you reset it — on average to a time about 30 seconds behind the actual time. so if you forgot to wind a 3919 every two weeks or so, it would rarely have shown the wrong time.

10 there's one brand still waiting to be reinflated: universal genève, which was one of the big players of the golden age but since 1977 has been little more than a brand name passed from acquirer to acquirer. they're scheduled to come back to life later this year, no doubt with stories about their long tradition of watchmaking.

11 more precisely, a high ratio of size to accuracy meant cheap. it's easier to make a larger movement keep good time, but between two watches of the same accuracy, the larger was usually the cheaper.

12 their form did once follow function. they were originally diving watches. but they're long since obsolete for this purpose. present day diving watches (now called dive computers) are digital and tell you much more than the time.

13 rolex was awarded an average of 16.6 patents per year in the 1950s, but only 1.7 per year in the 1960s.

pierre-yves donzé, *the making of a status symbol: a business history of rolex*, manchester university press, 2025.

14 rolexes also shared something more specific with suvs: aspirational manliness. an internal 1967 report by rolex's ad agency j. walter thompson explained the idea they were trying to convey: "because a rolex is designed for any situation, however rough or dangerous or heroic or exalted, it implies that the man who wears it is, potentially, a hero."

reprinted in donzé, *op cit*.

15 this business model only works when purchase decisions are driven mainly by brand. in a normal market, if one manufacturer restricts production, customers just buy from whichever competing manufacturer offers something as good. it's only when customers are seeking a certain brand rather than a certain level of performance that you can manipulate them by restricting its availability.

16 of course the first question one has on noticing a bubble is: will it burst? the reason ordinary bubbles eventually burst is that speculators get overoptimistic, but in this case the ceo of patek philippe controls the "money supply" and can thus take measures to cool down an overheated market. so there are probably only two things that could cause their specific bubble to burst: if his successor is not as capable, or if the whole custom of wearing mechanical watches goes away. the latter seems the greater danger. people aren't going to wear three things on their wrists, so all it would take is for there to be two popular devices that were worn on the wrist, and mechanical watches would start to be seen by the next cohort of young rich people as an old guy thing. it's hard to imagine a luxury watch brand surviving that.

thanks to sam altman, bill clerico, daniel gackle, luis garcia, the people at goldammer, jessica livingston, ben miller, robert morris, john reardon, d'arcy rice, alex tabarrok, and garry tan for reading drafts of this.